Supplementary Material for: Beyond Reconstruction: Approximating Task-Optimal Imputation via Surrogate Task Signals

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1 Theoretical Proofs

This appendix provides complete proofs for the theoretical results presented in Section 3 of the main paper.

1.1 Proof of Variance Reduction

Proposition (Variance Reduction, restated). For any random variables X, $\tilde{X}$, and Y,

$$\mathrm{Var}(X|\tilde{X}, Y) \leq \mathrm{Var}(X|\tilde{X}) \tag{S1}$$

with equality if and only if X and Y are conditionally independent given $\tilde{X}$.

Proof. We apply the law of total variance. For any random variables A and B, the law states that

$$\mathrm{Var}(A) = \mathbb{E}[\mathrm{Var}(A|B)] + \mathrm{Var}(\mathbb{E}[A|B]). \tag{S2}$$

Applying this to the conditional distribution of X given $\tilde{X}$, with Y as the additional conditioning variable, we obtain

$$\mathrm{Var}(X|\tilde{X}) = \mathbb{E}[\mathrm{Var}(X|\tilde{X}, Y)|\tilde{X}] + \mathrm{Var}(\mathbb{E}[X|\tilde{X}, Y]|\tilde{X}). \tag{S3}$$

Since variance is non-negative, the second term satisfies

$$\mathrm{Var}(\mathbb{E}[X|\tilde{X}, Y]|\tilde{X}) \geq 0. \tag{S4}$$

Therefore, from Equation (S3), we have

$$\mathrm{Var}(X|\tilde{X}) \geq \mathbb{E}[\mathrm{Var}(X|\tilde{X}, Y)|\tilde{X}]. \tag{S5}$$

To complete the proof, we note that for any fixed value of $\tilde{X}$, the conditional variance $\mathrm{Var}(X|\tilde{X}, Y)$ is a function of Y. Taking the expectation over Y given $\tilde{X}$ yields

$$\mathbb{E}[\mathrm{Var}(X|\tilde{X}, Y)|\tilde{X}] \geq \min_y \mathrm{Var}(X|\tilde{X}, Y = y). \tag{S6}$$

More directly, since we condition on more information in $\mathrm{Var}(X|\tilde{X}, Y)$ than in $\mathrm{Var}(X|\tilde{X})$, and the law of total variance shows that the additional variance term is non-negative, we conclude

$$\mathrm{Var}(X|\tilde{X}) \geq \mathrm{Var}(X|\tilde{X}, Y). \tag{S7}$$

Equality holds when $\mathrm{Var}(\mathbb{E}[X|\tilde{X}, Y]|\tilde{X}) = 0$, which occurs if and only if $\mathbb{E}[X|\tilde{X}, Y]$ does not depend on Y given $\tilde{X}$. This is equivalent to the condition that X and Y are conditionally independent given $\tilde{X}$, denoted $X \perp Y|\tilde{X}$. □

1.2 Proof of Proposition 3.1 (Task Signal Approximation)

Proposition (Pseudo-Label Approximation, restated). Let $\hat{Y}$ be pseudo-labels with accuracy $\eta = P(\hat{Y} = Y)$ for classification. Define $\Phi^{\hat{Y}}(\tilde{X}) = \mathbb{E}[X|\tilde{X}, \hat{Y}]$ as the pseudo-label conditioned imputation. Then

$$\mathbb{E}\left[\|\Phi^{\hat{Y}}(\tilde{X}) - \Phi^{TO}(\tilde{X})\|^2\right] \leq C \cdot (1-\eta) \tag{S8}$$

where $C = \max_{y \neq y'} \mathbb{E}\left[\|\mathbb{E}[X|\tilde{X}, Y{=}y] - \mathbb{E}[X|\tilde{X}, Y{=}y']\|^2\right]$ is the maximum squared distance between class-conditional imputations.

Proof. We decompose the expectation by conditioning on whether the pseudo-label equals the true label.

$$\begin{aligned}
&\mathbb{E}\left[\|\Phi^{\hat{Y}}(\tilde{X}) - \Phi^{TO}(\tilde{X})\|^2\right] \\
&= P(\hat{Y} = Y) \cdot \mathbb{E}\left[\|\Phi^{\hat{Y}} - \Phi^{TO}\|^2 \mid \hat{Y} = Y\right] \\
&\quad + P(\hat{Y} \neq Y) \cdot \mathbb{E}\left[\|\Phi^{\hat{Y}} - \Phi^{TO}\|^2 \mid \hat{Y} \neq Y\right].
\end{aligned} \tag{S9}$$

We analyze each term separately.

First term: When $\hat{Y} = Y$, the pseudo-label conditioned imputation equals the task-optimal imputation:

$$\Phi^{\hat{Y}}(\tilde{X}) = \mathbb{E}[X|\tilde{X}, \hat{Y}] = \mathbb{E}[X|\tilde{X}, Y] = \Phi^{TO}(\tilde{X}). \tag{S10}$$

Therefore, the first term vanishes:

$$\mathbb{E}\left[\|\Phi^{\hat{Y}} - \Phi^{TO}\|^2 \mid \hat{Y} = Y\right] = 0. \tag{S11}$$

Second term: When $\hat{Y} \neq Y$, we have a mismatch between the conditioning labels. The squared distance is

$$\begin{aligned}
&\|\Phi^{\hat{Y}}(\tilde{X}) - \Phi^{TO}(\tilde{X})\|^2 \\
&= \|\mathbb{E}[X|\tilde{X}, \hat{Y}] - \mathbb{E}[X|\tilde{X}, Y]\|^2.
\end{aligned} \tag{S12}$$

This quantity depends on the specific values of $\hat{Y}$ and Y. We bound it by the maximum over all mismatched pairs:

$$\begin{aligned}
&\mathbb{E}\left[\|\mathbb{E}[X|\tilde{X}, \hat{Y}] - \mathbb{E}[X|\tilde{X}, Y]\|^2 \mid \hat{Y} \neq Y\right] \\
&\leq \max_{y \neq y'} \mathbb{E}\left[\|\mathbb{E}[X|\tilde{X}, Y = y] - \mathbb{E}[X|\tilde{X}, Y = y']\|^2\right] \\
&= C.
\end{aligned} \tag{S13}$$

Combining terms: Substituting into Equation (S9) with $P(\hat{Y} = Y) = \eta$ and $P(\hat{Y} \neq Y) = 1 - \eta$:

$$\begin{aligned}
\mathbb{E}\left[\|\Phi^{\hat{Y}} - \Phi^{TO}\|^2\right] &= \eta \cdot 0 + (1-\eta) \cdot \mathbb{E}\left[\|\Phi^{\hat{Y}} - \Phi^{TO}\|^2 \mid \hat{Y} \neq Y\right] \\
&\leq (1-\eta) \cdot C \\
&= C \cdot (1-\eta).
\end{aligned} \tag{S14}$$

□

1.3 Extension to Regression Setting

For regression tasks where $Y \in \mathbb{R}$ is continuous, we establish an analogous result using a Lipschitz continuity assumption.

Proposition 1.1 (Pseudo-Label Approximation for Regression)**.** *Let $Y \in \mathbb{R}$ be a continuous label and assume the conditional expectation function $y \mapsto \mathbb{E}[X|\tilde{X}, Y = y]$ is L-Lipschitz, i.e.,*

$$\|\mathbb{E}[X|\tilde{X}, Y{=}y] - \mathbb{E}[X|\tilde{X}, Y{=}y']\| \leq L|y - y'| \tag{S15}$$

for all $y, y' \in \mathbb{R}$. Then for pseudo-labels $\hat{Y}$ with mean squared error $\sigma^2 = \mathbb{E}[(\hat{Y} - Y)^2]$,

$$\mathbb{E}\left[\|\Phi^{\hat{Y}}(\tilde{X}) - \Phi^{TO}(\tilde{X})\|^2\right] \leq L^2\sigma^2. \tag{S16}$$

Proof. By the Lipschitz assumption, for any realization of $\tilde{X}$, Y, and $\hat{Y}$:

$$\begin{aligned} \|\Phi^{\hat{Y}}(\tilde{X}) - \Phi^{TO}(\tilde{X})\| &= \|\mathbb{E}[X|\tilde{X}, \hat{Y}] - \mathbb{E}[X|\tilde{X}, Y]\| \\ &\leq L|\hat{Y} - Y|. \end{aligned} \tag{S17}$$

Squaring both sides:

$$\|\Phi^{\hat{Y}}(\tilde{X}) - \Phi^{TO}(\tilde{X})\|^2 \leq L^2|\hat{Y} - Y|^2. \tag{S18}$$

Taking expectations:

$$\mathbb{E}\left[\|\Phi^{\hat{Y}}(\tilde{X}) - \Phi^{TO}(\tilde{X})\|^2\right] \leq L^2 \cdot \mathbb{E}\left[(\hat{Y} - Y)^2\right] = L^2\sigma^2. \tag{S19}$$

□

Interpretation. The Lipschitz constant L measures the sensitivity of optimal imputation to label values. When L is small, the imputation is relatively insensitive to label errors, and even noisy pseudo-labels provide useful guidance. The bound guarantees that as pseudo-label quality improves (lower MSE σ^2), the imputation approaches the task-optimal solution.

Corollary 1.2 (Convergence)**.** *If the task-specific predictor is consistent, meaning $\sigma^2 = \mathbb{E}[(\hat{Y} - Y)^2] \to 0$ as $n \to \infty$, then*

$$\Phi^{\hat{Y}} \to \Phi^{TO} \quad \textit{in mean square.} \tag{S20}$$

This corollary establishes that TAIF asymptotically achieves task-optimal imputation as the amount of training data increases and pseudo-label quality improves.

1.4 Variance Reduction: Empirical Details

Table S1 presents the complete variance values for the variance reduction property discussed in the main paper. We empirically verify that conditioning on the label Y consistently reduces the variance of missing value imputation across all datasets.

Table S1: Empirical validation of the variance reduction property: Variance reduction through Y-conditioning. All datasets show positive variance reduction, confirming that $\mathrm{Var}(X|\tilde{X}, Y) \leq \mathrm{Var}(X|\tilde{X})$.

Dataset	**Task**	**Var**$(X\|\tilde{X})$	**Var**$(X\|\tilde{X}, Y)$	**Reduction (%)**	**FL Corr.**
Iris	Classification	0.081 ± 0.005	0.074 ± 0.004	11.4 ± 2.4	0.778
Wine	Classification	2562.8 ± 288.1	1878.9 ± 155.6	9.4 ± 1.7	0.482
Energy	Regression	0.461 ± 0.008	0.451 ± 0.007	8.4 ± 0.5	0.411
Kin8nm	Regression	0.819 ± 0.004	0.763 ± 0.005	6.8 ± 0.1	0.176
Housing	Regression	776.7 ± 45.4	759.1 ± 47.1	5.6 ± 0.7	0.205
Digits	Classification	0.019 ± 0.000	0.018 ± 0.000	0.9 ± 0.1	0.102

We define feature-label correlation (FL Corr.) as the mean absolute Spearman correlation between features and labels:

$$\text{FL Corr.} = \frac{1}{D} \sum_{j=1}^{D} |\rho_s(X_j, Y)|. \tag{S21}$$

Variance reduction strongly correlates with feature-label dependence (Spearman $\rho = 0.94$, $p < 0.05$). Iris exhibits the highest FL correlation (0.78) and largest reduction (11.4%), while Digits shows the lowest FL correlation (0.10) and minimal reduction (0.9%), suggesting near conditional independence $X \perp Y|\tilde{X}$.

The results confirm the variance reduction property across all six datasets. The magnitude of variance reduction varies substantially, ranging from 0.9% (Digits) to 11.4% (Iris). This variation reflects the strength of conditional dependence between features and labels: datasets where features strongly predict labels exhibit greater variance reduction when conditioning on Y.

The results demonstrate that the theoretical bound is satisfied in all configurations, validating the variance reduction property. Task signal accuracy η degrades gracefully with missing rate, averaging 0.43, yet the bound remains satisfied throughout. The empirical error consistently falls below the theoretical upper bound across all datasets, missing rates, and pseudo-label methods.

2 Full Regression Experimental Results

This appendix provides complete experimental results across all methods, missing rates, and evaluation metrics. We report both imputation quality (MAE on missing values) and downstream prediction performance (MAE with XGBoost predictor) for all regression datasets. Kin8nm values are scaled by $\times 100$ for readability.

Table S2: Imputation performance (MAE ↓) on test set across different missing rates. Best results within each base method in **bold**.

Base	Method	Missing Rate = 30%						Missing Rate = 50%						Missing Rate = 70%						Missing Rate = 90%					
		Hou.	Ene.	Yac.	Kin.	Pow.	Pro.	Hou.	Ene.	Yac.	Kin.	Pow.	Pro.	Hou.	Ene.	Yac.	Kin.	Pow.	Pro.	Hou.	Ene.	Yac.	Kin.	Pow.	Pro.
	Baseline	**0.18**$_{\pm.01}$	**0.31**$_{\pm.01}$	**0.21**$_{\pm.02}$	**25.12**$_{\pm.29}$	**0.17**$_{\pm.00}$	**0.09**$_{\pm.00}$	**0.19**$_{\pm.01}$	**0.31**$_{\pm.01}$	**0.21**$_{\pm.01}$	**25.09**$_{\pm.20}$	**0.17**$_{\pm.00}$	**0.10**$_{\pm.00}$	**0.19**$_{\pm.01}$	**0.31**$_{\pm.00}$	**0.22**$_{\pm.01}$	**25.08**$_{\pm.20}$	**0.17**$_{\pm.01}$	0.10$_{\pm.00}$	**0.22**$_{\pm.02}$	**0.31**$_{\pm.00}$	**0.23**$_{\pm.01}$	**25.12**$_{\pm.18}$	**0.18**$_{\pm.01}$	**0.11**$_{\pm.00}$
Mean	+ TAIF-C	**0.18**$_{\pm.01}$	**0.31**$_{\pm.01}$	**0.21**$_{\pm.02}$	**25.12**$_{\pm.29}$	**0.17**$_{\pm.00}$	**0.09**$_{\pm.00}$	**0.19**$_{\pm.01}$	**0.31**$_{\pm.01}$	**0.21**$_{\pm.01}$	25.09$_{\pm.20}$	**0.17**$_{\pm.00}$	0.10$_{\pm.00}$	**0.19**$_{\pm.01}$	**0.31**$_{\pm.00}$	**0.22**$_{\pm.01}$	25.08$_{\pm.20}$	**0.17**$_{\pm.01}$	0.10$_{\pm.00}$	**0.22**$_{\pm.02}$	**0.31**$_{\pm.00}$	**0.23**$_{\pm.01}$	25.12$_{\pm.18}$	**0.18**$_{\pm.01}$	0.11$_{\pm.00}$
	+ TAIF-N	1.05$_{\pm.07}$	0.96$_{\pm.08}$	0.76$_{\pm.12}$	25.30$_{\pm.29}$	4.00$_{\pm.36}$	0.11$_{\pm.01}$	1.07$_{\pm.04}$	1.14$_{\pm.16}$	0.69$_{\pm.12}$	25.23$_{\pm.22}$	3.97$_{\pm.37}$	0.10$_{\pm.00}$	1.14$_{\pm.08}$	1.31$_{\pm.14}$	0.82$_{\pm.10}$	25.19$_{\pm.22}$	5.40$_{\pm1.05}$	**0.09**$_{\pm.00}$	1.18$_{\pm.05}$	1.91$_{\pm.22}$	1.10$_{\pm.36}$	27.13$_{\pm.75}$	6.91$_{\pm1.32}$	0.12$_{\pm.01}$
	Baseline	**0.16**$_{\pm.01}$	**0.33**$_{\pm.01}$	**0.21**$_{\pm.02}$	**25.12**$_{\pm.29}$	**0.17**$_{\pm.00}$	**0.09**$_{\pm.00}$	**0.17**$_{\pm.01}$	**0.32**$_{\pm.00}$	**0.20**$_{\pm.02}$	**25.09**$_{\pm.20}$	**0.17**$_{\pm.00}$	**0.09**$_{\pm.00}$	**0.17**$_{\pm.01}$	**0.31**$_{\pm.01}$	**0.20**$_{\pm.01}$	**25.09**$_{\pm.21}$	**0.17**$_{\pm.01}$	0.10$_{\pm.00}$	**0.20**$_{\pm.02}$	**0.32**$_{\pm.00}$	**0.22**$_{\pm.01}$	**25.14**$_{\pm.17}$	**0.18**$_{\pm.01}$	**0.11**$_{\pm.00}$
Median	+ TAIF-C	**0.16**$_{\pm.01}$	**0.33**$_{\pm.01}$	**0.21**$_{\pm.02}$	**25.12**$_{\pm.29}$	**0.17**$_{\pm.00}$	0.09$_{\pm.00}$	**0.17**$_{\pm.01}$	**0.32**$_{\pm.00}$	**0.20**$_{\pm.02}$	25.09$_{\pm.20}$	**0.17**$_{\pm.00}$	**0.09**$_{\pm.00}$	**0.17**$_{\pm.01}$	**0.31**$_{\pm.01}$	**0.20**$_{\pm.01}$	25.09$_{\pm.21}$	**0.17**$_{\pm.01}$	0.10$_{\pm.00}$	**0.20**$_{\pm.02}$	**0.32**$_{\pm.00}$	**0.22**$_{\pm.01}$	25.14$_{\pm.17}$	**0.18**$_{\pm.01}$	0.11$_{\pm.00}$
	+ TAIF-N	1.14$_{\pm.13}$	0.91$_{\pm.13}$	0.95$_{\pm.22}$	25.30$_{\pm.32}$	3.87$_{\pm.42}$	0.10$_{\pm.00}$	1.07$_{\pm.10}$	0.90$_{\pm.16}$	0.78$_{\pm.19}$	25.23$_{\pm.20}$	3.89$_{\pm.23}$	0.10$_{\pm.01}$	1.06$_{\pm.16}$	1.09$_{\pm.20}$	0.86$_{\pm.13}$	25.20$_{\pm.23}$	5.44$_{\pm1.15}$	**0.09**$_{\pm.00}$	1.05$_{\pm.10}$	1.62$_{\pm.27}$	1.97$_{\pm.54}$	27.26$_{\pm.67}$	6.93$_{\pm1.41}$	0.12$_{\pm.02}$
	Baseline	**0.10**$_{\pm.01}$	0.20$_{\pm.02}$	**0.18**$_{\pm.02}$	26.6$_{\pm.2}$	0.13$_{\pm.00}$	**0.21**$_{\pm.00}$	0.14$_{\pm.01}$	0.23$_{\pm.01}$	**0.21**$_{\pm.01}$	26.8$_{\pm.2}$	0.14$_{\pm.00}$	**0.28**$_{\pm.00}$	0.16$_{\pm.01}$	0.26$_{\pm.01}$	**0.23**$_{\pm.01}$	26.6$_{\pm.2}$	**0.15**$_{\pm.01}$	**0.28**$_{\pm.00}$	0.22$_{\pm.02}$	**0.31**$_{\pm.01}$	**0.24**$_{\pm.01}$	**26.0**$_{\pm.2}$	**0.18**$_{\pm.01}$	**0.35**$_{\pm.00}$
KNN	+ TAIF-C	0.12$_{\pm.00}$	**0.17**$_{\pm.01}$	0.25$_{\pm.03}$	26.9$_{\pm.3}$	**0.12**$_{\pm.00}$	**0.17**$_{\pm.00}$	**0.13**$_{\pm.01}$	**0.19**$_{\pm.01}$	0.26$_{\pm.01}$	26.9$_{\pm.3}$	**0.13**$_{\pm.00}$	**0.38**$_{\pm.00}$	**0.14**$_{\pm.01}$	**0.24**$_{\pm.01}$	0.25$_{\pm.01}$	26.9$_{\pm.2}$	**0.15**$_{\pm.01}$	**0.49**$_{\pm.01}$	**0.20**$_{\pm.02}$	**0.30**$_{\pm.01}$	0.26$_{\pm.02}$	27.0$_{\pm.4}$	**0.18**$_{\pm.01}$	**0.51**$_{\pm.01}$
	+ TAIF-N	1.10$_{\pm.12}$	0.95$_{\pm.09}$	0.92$_{\pm.13}$	**25.9**$_{\pm.3}$	4.01$_{\pm.21}$	**0.51**$_{\pm.33}$	1.09$_{\pm.21}$	0.95$_{\pm.06}$	0.79$_{\pm.15}$	**25.8**$_{\pm.2}$	4.14$_{\pm.37}$	**0.44**$_{\pm.26}$	1.06$_{\pm.13}$	1.14$_{\pm.22}$	0.99$_{\pm.33}$	**25.7**$_{\pm.2}$	5.05$_{\pm1.16}$	**0.45**$_{\pm.25}$	1.03$_{\pm.15}$	1.47$_{\pm.25}$	2.15$_{\pm.54}$	**25.8**$_{\pm.3}$	6.69$_{\pm.62}$	**0.43**$_{\pm.13}$
	Baseline	**0.14**$_{\pm.01}$	0.16$_{\pm.01}$	**0.17**$_{\pm.02}$	**25.1**$_{\pm.3}$	0.12$_{\pm.00}$	**0.06**$_{\pm.00}$	**0.14**$_{\pm.01}$	0.21$_{\pm.01}$	**0.21**$_{\pm.02}$	27.6$_{\pm1.2}$	**0.13**$_{\pm.00}$	**0.11**$_{\pm.01}$	**0.17**$_{\pm.02}$	0.28$_{\pm.01}$	**0.22**$_{\pm.01}$	26.8$_{\pm.4}$	**0.15**$_{\pm.01}$	**0.28**$_{\pm.07}$	0.24$_{\pm.03}$	0.31$_{\pm.01}$	0.25$_{\pm.02}$	26.0$_{\pm.3}$	**0.18**$_{\pm.00}$	**0.50**$_{\pm.01}$
MICE	+ TAIF-C	**0.14**$_{\pm.01}$	**0.15**$_{\pm.02}$	**0.17**$_{\pm.02}$	26.7$_{\pm.5}$	**0.11**$_{\pm.00}$	**0.06**$_{\pm.00}$	**0.14**$_{\pm.00}$	**0.20**$_{\pm.01}$	**0.21**$_{\pm.02}$	**26.3**$_{\pm.2}$	**0.13**$_{\pm.00}$	**0.11**$_{\pm.01}$	**0.17**$_{\pm.02}$	**0.23**$_{\pm.01}$	**0.22**$_{\pm.01}$	**25.7**$_{\pm.3}$	**0.15**$_{\pm.01}$	**0.27**$_{\pm.06}$	**0.24**$_{\pm.02}$	**0.30**$_{\pm.01}$	**0.24**$_{\pm.02}$	**25.8**$_{\pm.2}$	**0.18**$_{\pm.01}$	**0.50**$_{\pm.01}$
	+ TAIF-N	1.11$_{\pm.09}$	1.10$_{\pm.06}$	1.04$_{\pm.14}$	25.3$_{\pm.3}$	3.89$_{\pm.25}$	**0.60**$_{\pm.53}$	1.17$_{\pm.13}$	1.25$_{\pm.09}$	0.96$_{\pm.17}$	27.0$_{\pm.9}$	3.92$_{\pm.39}$	**0.56**$_{\pm.32}$	1.17$_{\pm.12}$	1.50$_{\pm.14}$	1.23$_{\pm.39}$	26.5$_{\pm.4}$	4.64$_{\pm.98}$	**0.44**$_{\pm.20}$	1.18$_{\pm.14}$	1.90$_{\pm.23}$	2.21$_{\pm.36}$	**25.8**$_{\pm.3}$	6.79$_{\pm.79}$	**0.57**$_{\pm.27}$
	Baseline	**0.08**$_{\pm.01}$	**0.13**$_{\pm.02}$	**0.14**$_{\pm.02}$	26.3$_{\pm.3}$	**0.11**$_{\pm.00}$	**0.02**$_{\pm.00}$	**0.09**$_{\pm.00}$	0.17$_{\pm.01}$	**0.19**$_{\pm.02}$	28.6$_{\pm.3}$	0.14$_{\pm.00}$	**0.03**$_{\pm.00}$	0.13$_{\pm.01}$	**0.23**$_{\pm.01}$	**0.25**$_{\pm.02}$	29.9$_{\pm.3}$	0.16$_{\pm.01}$	0.05$_{\pm.00}$	0.25$_{\pm.03}$	**0.33**$_{\pm.02}$	0.32$_{\pm.05}$	30.0$_{\pm1.0}$	**0.19**$_{\pm.01}$	0.09$_{\pm.00}$
MissForest	+ TAIF-C	**0.07**$_{\pm.01}$	**0.13**$_{\pm.01}$	**0.14**$_{\pm.02}$	26.7$_{\pm.2}$	**0.11**$_{\pm.00}$	0.03$_{\pm.00}$	**0.09**$_{\pm.01}$	**0.16**$_{\pm.01}$	**0.18**$_{\pm.02}$	27.8$_{\pm.3}$	**0.13**$_{\pm.00}$	0.03$_{\pm.00}$	**0.12**$_{\pm.01}$	**0.22**$_{\pm.01}$	**0.25**$_{\pm.01}$	29.2$_{\pm.3}$	**0.15**$_{\pm.01}$	**0.05**$_{\pm.00}$	**0.22**$_{\pm.03}$	**0.32**$_{\pm.02}$	**0.31**$_{\pm.03}$	29.3$_{\pm.9}$	**0.18**$_{\pm.01}$	**0.09**$_{\pm.00}$
	+ TAIF-N	1.07$_{\pm.11}$	1.15$_{\pm.09}$	0.96$_{\pm.14}$	**25.9**$_{\pm.3}$	3.61$_{\pm.06}$	0.11$_{\pm.00}$	1.01$_{\pm.07}$	1.46$_{\pm.26}$	1.00$_{\pm.36}$	**26.1**$_{\pm.1}$	3.60$_{\pm.07}$	0.10$_{\pm.00}$	1.05$_{\pm.10}$	1.60$_{\pm.12}$	1.17$_{\pm.13}$	**26.1**$_{\pm.3}$	3.62$_{\pm.05}$	0.09$_{\pm.00}$	1.06$_{\pm.14}$	1.94$_{\pm.41}$	2.06$_{\pm.51}$	**26.6**$_{\pm.2}$	6.08$_{\pm.53}$	0.14$_{\pm.01}$
	Baseline	0.13$_{\pm.01}$	0.21$_{\pm.01}$	**0.25**$_{\pm.01}$	27.7$_{\pm.5}$	0.13$_{\pm.00}$	0.05$_{\pm.00}$	0.22$_{\pm.04}$	0.27$_{\pm.03}$	**0.27**$_{\pm.01}$	27.8$_{\pm.6}$	0.15$_{\pm.01}$	**0.06**$_{\pm.00}$	0.30$_{\pm.05}$	0.37$_{\pm.04}$	**0.37**$_{\pm.03}$	35.6$_{\pm1.1}$	**0.22**$_{\pm.02}$	0.15$_{\pm.01}$	**0.38**$_{\pm.09}$	0.45$_{\pm.02}$	0.39$_{\pm.04}$	39.5$_{\pm2.7}$	0.40$_{\pm.04}$	0.18$_{\pm.02}$
GAIN	+ TAIF-C	**0.12**$_{\pm.01}$	**0.19**$_{\pm.01}$	**0.25**$_{\pm.02}$	**27.0**$_{\pm.4}$	**0.12**$_{\pm.00}$	**0.05**$_{\pm.00}$	**0.20**$_{\pm.03}$	**0.22**$_{\pm.02}$	**0.27**$_{\pm.02}$	**26.6**$_{\pm.2}$	**0.14**$_{\pm.01}$	0.07$_{\pm.01}$	**0.29**$_{\pm.05}$	**0.36**$_{\pm.03}$	0.38$_{\pm.03}$	37.4$_{\pm2.2}$	0.31$_{\pm.02}$	0.14$_{\pm.01}$	0.38$_{\pm.09}$	**0.43**$_{\pm.02}$	**0.38**$_{\pm.07}$	37.9$_{\pm2.3}$	**0.27**$_{\pm.07}$	**0.17**$_{\pm.02}$
	+ TAIF-N	1.28$_{\pm.12}$	1.17$_{\pm.03}$	0.77$_{\pm.08}$	26.5$_{\pm.4}$	3.73$_{\pm.09}$	0.11$_{\pm.00}$	1.28$_{\pm.10}$	1.17$_{\pm.13}$	0.82$_{\pm.22}$	26.9$_{\pm.4}$	3.72$_{\pm.09}$	0.11$_{\pm.00}$	1.42$_{\pm.24}$	1.89$_{\pm.22}$	0.91$_{\pm.44}$	**27.8**$_{\pm.9}$	4.59$_{\pm.55}$	**0.12**$_{\pm.02}$	1.31$_{\pm.42}$	2.48$_{\pm.39}$	1.75$_{\pm1.29}$	**28.5**$_{\pm.6}$	6.66$_{\pm.52}$	0.20$_{\pm.05}$
	Baseline	**0.51**$_{\pm.03}$	**0.35**$_{\pm.01}$	0.39$_{\pm.02}$	26.0$_{\pm.4}$	0.18$_{\pm.00}$	0.10$_{\pm.00}$	**0.50**$_{\pm.02}$	**0.37**$_{\pm.02}$	0.39$_{\pm.02}$	26.1$_{\pm.2}$	0.18$_{\pm.00}$	0.10$_{\pm.00}$	**0.51**$_{\pm.01}$	**0.38**$_{\pm.01}$	0.40$_{\pm.02}$	26.1$_{\pm.2}$	0.18$_{\pm.01}$	0.10$_{\pm.00}$	**0.54**$_{\pm.02}$	**0.39**$_{\pm.01}$	0.44$_{\pm.03}$	26.2$_{\pm.2}$	**0.19**$_{\pm.01}$	**0.11**$_{\pm.00}$
MIWAE	+ TAIF-C	0.54$_{\pm.02}$	0.39$_{\pm.02}$	**0.30**$_{\pm.02}$	28.1$_{\pm.3}$	**0.14**$_{\pm.00}$	**0.06**$_{\pm.00}$	0.53$_{\pm.01}$	0.39$_{\pm.02}$	**0.29**$_{\pm.01}$	28.0$_{\pm.2}$	**0.15**$_{\pm.00}$	**0.06**$_{\pm.00}$	0.53$_{\pm.01}$	0.39$_{\pm.01}$	**0.30**$_{\pm.01}$	26.6$_{\pm.7}$	**0.17**$_{\pm.01}$	**0.07**$_{\pm.00}$	0.56$_{\pm.01}$	**0.38**$_{\pm.01}$	**0.32**$_{\pm.02}$	28.4$_{\pm.8}$	0.19$_{\pm.01}$	0.11$_{\pm.00}$
	+ TAIF-N	1.33$_{\pm.08}$	0.97$_{\pm.07}$	1.50$_{\pm.48}$	**25.9**$_{\pm.4}$	4.03$_{\pm.07}$	0.10$_{\pm.00}$	1.50$_{\pm.15}$	1.09$_{\pm.09}$	1.42$_{\pm.41}$	**25.8**$_{\pm.2}$	3.93$_{\pm.07}$	0.09$_{\pm.00}$	1.65$_{\pm.20}$	1.37$_{\pm.14}$	1.57$_{\pm.32}$	**25.7**$_{\pm.2}$	6.86$_{\pm.42}$	0.10$_{\pm.00}$	1.41$_{\pm.23}$	1.56$_{\pm.17}$	1.86$_{\pm.48}$	**25.6**$_{\pm.1}$	7.70$_{\pm.41}$	0.14$_{\pm.01}$
	Baseline	0.15$_{\pm.01}$	0.25$_{\pm.01}$	0.31$_{\pm.03}$	29.03$_{\pm.23}$	0.20$_{\pm.00}$	**0.07**$_{\pm.00}$	0.16$_{\pm.01}$	0.29$_{\pm.01}$	0.38$_{\pm.10}$	31.47$_{\pm.26}$	0.26$_{\pm.00}$	0.08$_{\pm.00}$	0.25$_{\pm.07}$	0.36$_{\pm.02}$	0.45$_{\pm.07}$	35.80$_{\pm.36}$	0.35$_{\pm.00}$	0.12$_{\pm.01}$	0.40$_{\pm.02}$	0.52$_{\pm.01}$	0.52$_{\pm.02}$	45.07$_{\pm2.31}$	0.47$_{\pm.01}$	0.20$_{\pm.01}$
SoftImpute	+ TAIF-C	**0.14**$_{\pm.01}$	**0.23**$_{\pm.01}$	**0.29**$_{\pm.04}$	27.49$_{\pm.25}$	**0.17**$_{\pm.00}$	0.07$_{\pm.01}$	**0.14**$_{\pm.01}$	**0.25**$_{\pm.01}$	**0.31**$_{\pm.06}$	28.01$_{\pm.34}$	**0.17**$_{\pm.00}$	**0.07**$_{\pm.01}$	**0.18**$_{\pm.04}$	**0.27**$_{\pm.02}$	**0.33**$_{\pm.04}$	28.52$_{\pm.26}$	**0.18**$_{\pm.01}$	**0.08**$_{\pm.00}$	**0.30**$_{\pm.05}$	**0.34**$_{\pm.02}$	**0.37**$_{\pm.05}$	**31.12**$_{\pm.64}$	**0.19**$_{\pm.01}$	**0.09**$_{\pm.00}$
	+ TAIF-N	1.05$_{\pm.07}$	0.98$_{\pm.06}$	0.81$_{\pm.10}$	**26.82**$_{\pm.38}$	4.59$_{\pm.12}$	0.09$_{\pm.00}$	1.16$_{\pm.08}$	1.13$_{\pm.19}$	0.68$_{\pm.15}$	**27.42**$_{\pm.28}$	4.64$_{\pm.07}$	0.09$_{\pm.00}$	1.17$_{\pm.14}$	1.32$_{\pm.10}$	0.71$_{\pm.24}$	**27.93**$_{\pm.42}$	5.51$_{\pm.27}$	0.09$_{\pm.00}$	1.23$_{\pm.11}$	1.81$_{\pm.29}$	1.02$_{\pm.30}$	34.27$_{\pm.57}$	8.13$_{\pm.49}$	0.11$_{\pm.01}$
	Baseline	**0.14**$_{\pm.01}$	**0.26**$_{\pm.01}$	**0.21**$_{\pm.02}$	**25.27**$_{\pm.31}$	0.15$_{\pm.01}$	**0.07**$_{\pm.00}$	0.17$_{\pm.01}$	**0.28**$_{\pm.01}$	**0.22**$_{\pm.01}$	**25.21**$_{\pm.24}$	0.16$_{\pm.00}$	**0.08**$_{\pm.00}$	0.18$_{\pm.01}$	**0.30**$_{\pm.01}$	**0.22**$_{\pm.01}$	**25.09**$_{\pm.14}$	0.17$_{\pm.01}$	**0.09**$_{\pm.00}$	0.22$_{\pm.02}$	**0.31**$_{\pm.00}$	**0.23**$_{\pm.01}$	**25.12**$_{\pm.08}$	0.19$_{\pm.01}$	**0.11**$_{\pm.00}$
DiffPuter	+ TAIF-C	0.15$_{\pm.01}$	**0.21**$_{\pm.01}$	0.23$_{\pm.03}$	**25.27**$_{\pm.32}$	0.15$_{\pm.00}$	**0.06**$_{\pm.00}$	**0.16**$_{\pm.01}$	**0.25**$_{\pm.01}$	**0.22**$_{\pm.01}$	25.27$_{\pm.21}$	0.16$_{\pm.00}$	**0.08**$_{\pm.00}$	**0.18**$_{\pm.01}$	**0.28**$_{\pm.01}$	**0.22**$_{\pm.01}$	25.12$_{\pm.16}$	0.18$_{\pm.01}$	**0.09**$_{\pm.00}$	**0.22**$_{\pm.02}$	**0.30**$_{\pm.00}$	**0.23**$_{\pm.01}$	25.11$_{\pm.08}$	0.20$_{\pm.01}$	**0.11**$_{\pm.00}$
	+ TAIF-N	0.43$_{\pm.08}$	0.37$_{\pm.05}$	1.69$_{\pm.26}$	**25.27**$_{\pm.36}$	5.72$_{\pm2.85}$	**0.07**$_{\pm.00}$	0.33$_{\pm.05}$	0.31$_{\pm.06}$	1.19$_{\pm.20}$	25.23$_{\pm.20}$	5.32$_{\pm1.27}$	**0.08**$_{\pm.00}$	0.28$_{\pm.02}$	0.33$_{\pm.03}$	0.58$_{\pm.12}$	25.09$_{\pm.15}$	4.17$_{\pm.64}$	**0.09**$_{\pm.00}$	0.26$_{\pm.02}$	0.33$_{\pm.01}$	0.40$_{\pm.05}$	25.11$_{\pm.10}$	4.88$_{\pm1.46}$	**0.11**$_{\pm.00}$
	Baseline	**0.09**$_{\pm.00}$	**0.16**$_{\pm.02}$	**0.20**$_{\pm.02}$	**0.26**$_{\pm.00}$	**0.12**$_{\pm.01}$	**0.03**$_{\pm.00}$	**0.10**$_{\pm.00}$	**0.20**$_{\pm.01}$	**0.24**$_{\pm.01}$	0.27$_{\pm.00}$	**0.12**$_{\pm.00}$	**0.04**$_{\pm.00}$	**0.13**$_{\pm.00}$	0.26$_{\pm.01}$	0.28$_{\pm.02}$	0.27$_{\pm.01}$	**0.15**$_{\pm.01}$	**0.05**$_{\pm.00}$	0.23$_{\pm.02}$	0.33$_{\pm.01}$	0.34$_{\pm.02}$	0.27$_{\pm.01}$	**0.18**$_{\pm.01}$	**0.08**$_{\pm.00}$
GRAPE	+ TAIF-C	0.15$_{\pm.04}$	0.21$_{\pm.03}$	0.24$_{\pm.02}$	**0.26**$_{\pm.01}$	0.18$_{\pm.04}$	0.06$_{\pm.01}$	0.17$_{\pm.02}$	0.23$_{\pm.02}$	0.25$_{\pm.02}$	**0.26**$_{\pm.01}$	0.19$_{\pm.06}$	0.07$_{\pm.01}$	0.17$_{\pm.02}$	0.27$_{\pm.02}$	0.26$_{\pm.01}$	**0.26**$_{\pm.01}$	0.20$_{\pm.05}$	0.08$_{\pm.01}$	**0.21**$_{\pm.01}$	**0.31**$_{\pm.01}$	**0.27**$_{\pm.03}$	**0.26**$_{\pm.00}$	0.20$_{\pm.02}$	0.10$_{\pm.02}$
	+ TAIF-N	0.13$_{\pm.01}$	0.19$_{\pm.01}$	0.23$_{\pm.02}$	0.27$_{\pm.01}$	0.18$_{\pm.02}$	0.04$_{\pm.00}$	0.13$_{\pm.01}$	0.21$_{\pm.01}$	**0.24**$_{\pm.01}$	0.27$_{\pm.01}$	0.19$_{\pm.02}$	0.04$_{\pm.00}$	0.14$_{\pm.01}$	**0.24**$_{\pm.01}$	**0.25**$_{\pm.01}$	0.28$_{\pm.01}$	0.20$_{\pm.02}$	0.05$_{\pm.00}$	**0.21**$_{\pm.02}$	**0.31**$_{\pm.01}$	0.33$_{\pm.02}$	0.27$_{\pm.01}$	0.19$_{\pm.01}$	0.08$_{\pm.00}$

Table S3: Downstream prediction performance (MAE ↓) with XGBoost predictor across different missing rates. Best results within each base method in **bold**.

Base	Method	Missing Rate = 30%						Missing Rate = 50%						Missing Rate = 70%						Missing Rate = 90%					
		Hou.	Ene.	Yac.	Kin.	Pow.	Pro.	Hou.	Ene.	Yac.	Kin.	Pow.	Pro.	Hou.	Ene.	Yac.	Kin.	Pow.	Pro.	Hou.	Ene.	Yac.	Kin.	Pow.	Pro.
	Baseline	**3.48**$_{\pm.28}$	**1.52**$_{\pm.11}$	4.77$_{\pm1.03}$	**16.68**$_{\pm.25}$	**4.44**$_{\pm.11}$	3.94$_{\pm.01}$	**4.29**$_{\pm.21}$	**2.64**$_{\pm.16}$	8.17$_{\pm1.54}$	**18.92**$_{\pm.12}$	**6.27**$_{\pm.15}$	4.43$_{\pm.02}$	**5.25**$_{\pm.42}$	**4.27**$_{\pm.24}$	9.59$_{\pm.78}$	**20.53**$_{\pm.12}$	**8.98**$_{\pm.20}$	4.91$_{\pm.02}$	**6.19**$_{\pm.33}$	**7.10**$_{\pm.36}$	**10.49**$_{\pm1.29}$	**21.28**$_{\pm.15}$	**12.58**$_{\pm.16}$	5.34$_{\pm.01}$
Mean	+ TAIF-C	**3.48**$_{\pm.28}$	**1.52**$_{\pm.11}$	4.77$_{\pm1.03}$	16.68$_{\pm.25}$	**4.44**$_{\pm.11}$	3.94$_{\pm.01}$	**4.29**$_{\pm.21}$	**2.64**$_{\pm.16}$	8.17$_{\pm1.54}$	18.92$_{\pm.12}$	**6.27**$_{\pm.15}$	4.43$_{\pm.02}$	**5.25**$_{\pm.42}$	**4.27**$_{\pm.24}$	9.59$_{\pm.78}$	20.53$_{\pm.12}$	**8.98**$_{\pm.20}$	4.91$_{\pm.02}$	**6.19**$_{\pm.33}$	**7.10**$_{\pm.36}$	**10.49**$_{\pm1.29}$	21.28$_{\pm.15}$	**12.58**$_{\pm.16}$	5.34$_{\pm.01}$
	+ TAIF-N	3.90$_{\pm.42}$	2.19$_{\pm.20}$	**3.86**$_{\pm.81}$	16.91$_{\pm.19}$	**4.44**$_{\pm.13}$	**3.87**$_{\pm.03}$	5.09$_{\pm.53}$	3.50$_{\pm.27}$	**7.08**$_{\pm1.15}$	19.08$_{\pm.17}$	6.29$_{\pm.16}$	**4.30**$_{\pm.03}$	5.88$_{\pm.46}$	4.83$_{\pm.41}$	**8.88**$_{\pm1.04}$	20.85$_{\pm.17}$	**8.98**$_{\pm.20}$	**4.78**$_{\pm.02}$	6.61$_{\pm.53}$	7.37$_{\pm.33}$	10.85$_{\pm1.26}$	21.59$_{\pm.16}$	12.66$_{\pm.27}$	**5.32**$_{\pm.01}$
	Baseline	**3.44**$_{\pm.33}$	**1.68**$_{\pm.19}$	5.51$_{\pm1.26}$	**16.64**$_{\pm.29}$	4.45$_{\pm.12}$	3.94$_{\pm.01}$	**4.27**$_{\pm.38}$	**2.92**$_{\pm.31}$	7.88$_{\pm1.09}$	**18.88**$_{\pm.15}$	6.29$_{\pm.17}$	4.42$_{\pm.03}$	**5.25**$_{\pm.51}$	**4.90**$_{\pm.25}$	9.45$_{\pm.85}$	**20.50**$_{\pm.12}$	**8.97**$_{\pm.19}$	4.90$_{\pm.01}$	**6.21**$_{\pm.22}$	7.60$_{\pm.34}$	**10.44**$_{\pm1.10}$	**21.29**$_{\pm.17}$	**12.59**$_{\pm.16}$	5.34$_{\pm.01}$
Median	+ TAIF-C	**3.44**$_{\pm.33}$	**1.68**$_{\pm.19}$	5.51$_{\pm1.26}$	16.64$_{\pm.29}$	4.45$_{\pm.12}$	3.94$_{\pm.01}$	**4.27**$_{\pm.38}$	**2.92**$_{\pm.31}$	7.88$_{\pm1.09}$	18.88$_{\pm.15}$	6.29$_{\pm.17}$	4.42$_{\pm.03}$	**5.25**$_{\pm.51}$	**4.90**$_{\pm.25}$	9.45$_{\pm.85}$	20.50$_{\pm.12}$	**8.97**$_{\pm.19}$	4.90$_{\pm.01}$	**6.21**$_{\pm.22}$	7.60$_{\pm.34}$	**10.44**$_{\pm1.10}$	21.29$_{\pm.17}$	**12.59**$_{\pm.16}$	5.34$_{\pm.01}$
	+ TAIF-N	4.05$_{\pm.42}$	2.05$_{\pm.17}$	**4.43**$_{\pm1.24}$	16.93$_{\pm.21}$	**4.44**$_{\pm.12}$	**3.88**$_{\pm.04}$	5.05$_{\pm.50}$	3.47$_{\pm.31}$	**7.13**$_{\pm1.28}$	19.11$_{\pm.17}$	**6.26**$_{\pm.16}$	**4.29**$_{\pm.04}$	6.10$_{\pm.43}$	4.97$_{\pm.36}$	**8.98**$_{\pm.80}$	20.76$_{\pm.19}$	9.02$_{\pm.20}$	**4.78**$_{\pm.03}$	6.53$_{\pm.62}$	**7.41**$_{\pm.29}$	10.91$_{\pm1.13}$	21.53$_{\pm.19}$	**12.59**$_{\pm.15}$	**5.32**$_{\pm.03}$
	Baseline	3.36$_{\pm.25}$	1.67$_{\pm.18}$	4.50$_{\pm1.04}$	17.32$_{\pm.16}$	4.83$_{\pm.18}$	**4.02**$_{\pm.03}$	4.61$_{\pm.33}$	3.13$_{\pm.31}$	7.54$_{\pm1.22}$	19.43$_{\pm.24}$	6.70$_{\pm.12}$	**4.65**$_{\pm.01}$	5.65$_{\pm.52}$	4.85$_{\pm.25}$	9.46$_{\pm1.26}$	20.81$_{\pm.20}$	9.35$_{\pm.16}$	**5.08**$_{\pm.02}$	6.43$_{\pm.26}$	7.27$_{\pm.26}$	10.62$_{\pm1.33}$	21.50$_{\pm.17}$	**12.69**$_{\pm.15}$	**5.37**$_{\pm.01}$
KNN	+ TAIF-C	**3.11**$_{\pm.35}$	**1.37**$_{\pm.18}$	4.31$_{\pm1.01}$	17.42$_{\pm.22}$	**4.41**$_{\pm.12}$	**3.76**$_{\pm.02}$	**3.92**$_{\pm.48}$	**2.47**$_{\pm.18}$	**6.99**$_{\pm1.27}$	19.94$_{\pm.34}$	6.38$_{\pm.18}$	4.34$_{\pm.03}$	**4.82**$_{\pm.37}$	**4.41**$_{\pm.26}$	**9.07**$_{\pm1.10}$	22.06$_{\pm.33}$	9.25$_{\pm.25}$	4.82$_{\pm.02}$	**6.34**$_{\pm.32}$	**7.21**$_{\pm.34}$	**10.60**$_{\pm1.34}$	22.97$_{\pm.26}$	12.86$_{\pm.18}$	5.38$_{\pm.02}$
	+ TAIF-N	3.80$_{\pm.29}$	2.03$_{\pm.21}$	**4.22**$_{\pm1.16}$	**16.96**$_{\pm.26}$	4.44$_{\pm.14}$	**3.88**$_{\pm.05}$	5.19$_{\pm.48}$	3.49$_{\pm.37}$	7.81$_{\pm1.04}$	**19.22**$_{\pm.20}$	**6.25**$_{\pm.16}$	**4.29**$_{\pm.03}$	6.21$_{\pm.57}$	4.99$_{\pm.28}$	9.51$_{\pm.90}$	**20.69**$_{\pm.17}$	**9.05**$_{\pm.23}$	**4.81**$_{\pm.03}$	6.64$_{\pm.37}$	7.50$_{\pm.29}$	10.80$_{\pm1.13}$	**21.63**$_{\pm.16}$	13.11$_{\pm1.26}$	**5.33**$_{\pm.03}$
	Baseline	3.68$_{\pm.26}$	1.39$_{\pm.18}$	3.97$_{\pm.90}$	**16.69**$_{\pm.20}$	4.50$_{\pm.13}$	**3.76**$_{\pm.02}$	4.25$_{\pm.65}$	2.69$_{\pm.18}$	7.74$_{\pm1.27}$	19.32$_{\pm.40}$	6.37$_{\pm.16}$	**4.28**$_{\pm.03}$	5.52$_{\pm.51}$	4.52$_{\pm.25}$	**9.40**$_{\pm1.03}$	20.85$_{\pm.13}$	9.17$_{\pm.20}$	**4.88**$_{\pm.01}$	6.52$_{\pm.43}$	7.29$_{\pm.36}$	10.79$_{\pm1.24}$	**21.49**$_{\pm.18}$	**12.65**$_{\pm.15}$	**5.33**$_{\pm.01}$
MICE	+ TAIF-C	**3.51**$_{\pm.27}$	**1.37**$_{\pm.16}$	3.84$_{\pm1.08}$	17.32$_{\pm.22}$	4.46$_{\pm.14}$	**3.65**$_{\pm.03}$	**4.11**$_{\pm.37}$	**2.64**$_{\pm.28}$	**7.44**$_{\pm1.16}$	19.78$_{\pm.27}$	6.51$_{\pm.18}$	4.09$_{\pm.04}$	**5.16**$_{\pm.51}$	**4.42**$_{\pm.27}$	9.58$_{\pm.75}$	22.23$_{\pm.26}$	9.41$_{\pm.26}$	4.71$_{\pm.06}$	**6.34**$_{\pm.31}$	**7.21**$_{\pm.36}$	**10.76**$_{\pm1.32}$	23.53$_{\pm.26}$	12.99$_{\pm.21}$	5.50$_{\pm.04}$
	+ TAIF-N	3.83$_{\pm.43}$	1.95$_{\pm.13}$	**3.70**$_{\pm.91}$	16.84$_{\pm.25}$	**4.44**$_{\pm.14}$	**3.82**$_{\pm.03}$	4.96$_{\pm.45}$	3.33$_{\pm.25}$	7.46$_{\pm1.17}$	**19.22**$_{\pm.23}$	**6.26**$_{\pm.15}$	**4.24**$_{\pm.03}$	6.00$_{\pm.41}$	4.92$_{\pm.16}$	9.50$_{\pm1.21}$	**20.76**$_{\pm.15}$	**9.00**$_{\pm.18}$	**4.77**$_{\pm.03}$	6.68$_{\pm.33}$	7.40$_{\pm.36}$	10.97$_{\pm1.30}$	21.53$_{\pm.16}$	13.37$_{\pm1.37}$	**5.31**$_{\pm.02}$
	Baseline	3.35$_{\pm.33}$	1.31$_{\pm.19}$	4.20$_{\pm1.06}$	17.14$_{\pm.22}$	4.52$_{\pm.12}$	3.59$_{\pm.01}$	4.34$_{\pm.36}$	2.84$_{\pm.29}$	7.79$_{\pm1.12}$	19.68$_{\pm.16}$	6.61$_{\pm.15}$	4.15$_{\pm.02}$	5.59$_{\pm.57}$	4.79$_{\pm.24}$	9.80$_{\pm1.18}$	21.07$_{\pm.17}$	9.37$_{\pm.22}$	4.81$_{\pm.03}$	6.59$_{\pm.29}$	7.53$_{\pm.50}$	10.92$_{\pm1.40}$	21.71$_{\pm.22}$	12.87$_{\pm.23}$	5.38$_{\pm.02}$
MissForest	+ TAIF-C	**3.08**$_{\pm.31}$	**1.25**$_{\pm.13}$	**4.18**$_{\pm.97}$	17.15$_{\pm.21}$	**4.36**$_{\pm.14}$	**3.53**$_{\pm.01}$	**3.93**$_{\pm.44}$	**2.44**$_{\pm.19}$	**7.04**$_{\pm1.26}$	19.84$_{\pm.21}$	6.38$_{\pm.18}$	**4.06**$_{\pm.01}$	**4.78**$_{\pm.33}$	**4.37**$_{\pm.38}$	**9.13**$_{\pm1.08}$	22.26$_{\pm.26}$	9.30$_{\pm.24}$	**4.72**$_{\pm.02}$	**6.26**$_{\pm.48}$	**7.18**$_{\pm.33}$	**10.63**$_{\pm1.33}$	23.11$_{\pm.22}$	**12.85**$_{\pm.17}$	5.38$_{\pm.03}$
	+ TAIF-N	3.63$_{\pm.35}$	1.80$_{\pm.14}$	4.24$_{\pm1.07}$	**16.96**$_{\pm.26}$	4.46$_{\pm.13}$	3.76$_{\pm.04}$	4.67$_{\pm.51}$	2.90$_{\pm.20}$	7.87$_{\pm1.36}$	**19.15**$_{\pm.20}$	**6.28**$_{\pm.16}$	4.22$_{\pm.04}$	5.82$_{\pm.50}$	4.66$_{\pm.30}$	9.68$_{\pm.97}$	**20.76**$_{\pm.18}$	**9.05**$_{\pm.20}$	4.76$_{\pm.02}$	6.54$_{\pm.41}$	7.41$_{\pm.30}$	10.87$_{\pm1.34}$	**21.56**$_{\pm.23}$	12.98$_{\pm.82}$	**5.33**$_{\pm.02}$
	Baseline	3.87$_{\pm.33}$	2.07$_{\pm.21}$	4.28$_{\pm1.14}$	17.70$_{\pm.22}$	4.81$_{\pm.15}$	4.06$_{\pm.04}$	7.29$_{\pm2.09}$	3.87$_{\pm.69}$	7.44$_{\pm1.16}$	20.07$_{\pm.39}$	7.41$_{\pm.66}$	**4.52**$_{\pm.05}$	8.54$_{\pm1.73}$	6.37$_{\pm.53}$	10.34$_{\pm1.74}$	22.76$_{\pm1.34}$	13.49$_{\pm1.21}$	5.16$_{\pm.05}$	9.58$_{\pm3.87}$	8.78$_{\pm.70}$	14.27$_{\pm5.50}$	22.73$_{\pm.56}$	16.28$_{\pm1.40}$	5.95$_{\pm.71}$
GAIN	+ TAIF-C	**3.45**$_{\pm.35}$	**1.69**$_{\pm.21}$	**4.13**$_{\pm1.25}$	**17.37**$_{\pm.27}$	4.60$_{\pm.15}$	3.99$_{\pm.13}$	**5.74**$_{\pm1.77}$	**3.15**$_{\pm.43}$	**6.14**$_{\pm1.42}$	20.29$_{\pm.54}$	7.53$_{\pm.64}$	4.53$_{\pm.16}$	**7.37**$_{\pm2.40}$	**5.67**$_{\pm.57}$	10.10$_{\pm2.98}$	24.14$_{\pm1.23}$	16.29$_{\pm2.99}$	5.91$_{\pm.51}$	**8.09**$_{\pm1.30}$	9.16$_{\pm.73}$	14.15$_{\pm8.32}$	25.56$_{\pm3.10}$	16.38$_{\pm4.51}$	5.83$_{\pm.73}$
	+ TAIF-N	4.01$_{\pm.40}$	2.11$_{\pm.18}$	4.47$_{\pm1.17}$	17.60$_{\pm.20}$	**4.50**$_{\pm.11}$	**3.95**$_{\pm.05}$	6.10$_{\pm.83}$	3.60$_{\pm.26}$	7.56$_{\pm1.39}$	**20.24**$_{\pm.17}$	**6.57**$_{\pm.27}$	4.54$_{\pm.06}$	9.20$_{\pm2.32}$	5.86$_{\pm.68}$	**9.95**$_{\pm3.01}$	**21.99**$_{\pm.62}$	**9.48**$_{\pm.41}$	**5.03**$_{\pm.15}$	9.50$_{\pm4.47}$	**8.19**$_{\pm.86}$	**10.27**$_{\pm3.48}$	**22.43**$_{\pm.54}$	**13.09**$_{\pm.51}$	**5.78**$_{\pm.18}$
	Baseline	**4.29**$_{\pm.56}$	2.94$_{\pm.20}$	4.49$_{\pm1.10}$	16.92$_{\pm.26}$	5.06$_{\pm.12}$	4.17$_{\pm.02}$	6.28$_{\pm1.04}$	4.98$_{\pm.39}$	7.28$_{\pm.90}$	19.28$_{\pm.25}$	7.14$_{\pm.18}$	4.68$_{\pm.04}$	7.89$_{\pm.98}$	7.05$_{\pm.63}$	10.29$_{\pm1.61}$	20.75$_{\pm.17}$	9.87$_{\pm.16}$	5.10$_{\pm.02}$	8.50$_{\pm1.38}$	9.26$_{\pm.69}$	12.36$_{\pm1.48}$	21.87$_{\pm.22}$	13.26$_{\pm.18}$	5.41$_{\pm.01}$
MIWAE	+ TAIF-C	4.86$_{\pm.52}$	3.08$_{\pm.42}$	**3.93**$_{\pm1.03}$	17.35$_{\pm.30}$	5.07$_{\pm.12}$	4.30$_{\pm.02}$	**6.17**$_{\pm.73}$	4.74$_{\pm.30}$	**6.68**$_{\pm1.39}$	19.71$_{\pm.22}$	7.11$_{\pm.14}$	4.86$_{\pm.02}$	**7.37**$_{\pm.76}$	6.66$_{\pm.33}$	**9.95**$_{\pm1.70}$	20.94$_{\pm.30}$	9.89$_{\pm.09}$	5.27$_{\pm.06}$	**8.28**$_{\pm.92}$	8.74$_{\pm.43}$	12.50$_{\pm2.15}$	22.44$_{\pm.34}$	13.62$_{\pm.12}$	5.41$_{\pm.01}$
	+ TAIF-N	5.44$_{\pm.50}$	**2.34**$_{\pm.16}$	6.07$_{\pm1.69}$	**16.86**$_{\pm.31}$	**4.44**$_{\pm.11}$	**3.90**$_{\pm.04}$	9.29$_{\pm1.34}$	**4.32**$_{\pm.38}$	10.73$_{\pm1.90}$	**19.18**$_{\pm.26}$	**6.26**$_{\pm.16}$	**4.34**$_{\pm.03}$	11.83$_{\pm1.88}$	**6.18**$_{\pm.60}$	12.58$_{\pm2.00}$	**20.69**$_{\pm.13}$	**9.06**$_{\pm.17}$	**4.82**$_{\pm.02}$	11.91$_{\pm3.30}$	**8.38**$_{\pm.44}$	**12.15**$_{\pm2.44}$	**21.80**$_{\pm.26}$	**12.68**$_{\pm.19}$	**5.34**$_{\pm.02}$
	Baseline	3.97$_{\pm.43}$	1.95$_{\pm.14}$	4.70$_{\pm1.49}$	17.67$_{\pm.26}$	5.07$_{\pm.14}$	3.97$_{\pm.01}$	4.84$_{\pm.32}$	3.30$_{\pm.27}$	7.83$_{\pm1.71}$	20.01$_{\pm.33}$	7.23$_{\pm.22}$	4.48$_{\pm.02}$	6.73$_{\pm.97}$	5.43$_{\pm.40}$	9.58$_{\pm1.32}$	21.89$_{\pm.67}$	9.83$_{\pm.20}$	5.02$_{\pm.05}$	**7.59**$_{\pm1.00}$	8.50$_{\pm.58}$	10.79$_{\pm1.36}$	22.47$_{\pm.78}$	13.85$_{\pm.50}$	5.39$_{\pm.02}$
SoftImpute	+ TAIF-C	**3.42**$_{\pm.35}$	**1.55**$_{\pm.15}$	**3.62**$_{\pm1.15}$	17.76$_{\pm.23}$	7.29$_{\pm.83}$	**3.80**$_{\pm.03}$	**4.06**$_{\pm.47}$	**2.76**$_{\pm.17}$	**7.32**$_{\pm1.33}$	21.87$_{\pm.54}$	12.12$_{\pm1.42}$	4.65$_{\pm.06}$	**5.29**$_{\pm.39}$	**4.53**$_{\pm.36}$	**9.06**$_{\pm1.34}$	24.11$_{\pm1.11}$	18.06$_{\pm2.64}$	5.31$_{\pm.29}$	7.83$_{\pm2.59}$	**7.75**$_{\pm.34}$	12.34$_{\pm1.86}$	25.65$_{\pm2.34}$	13.89$_{\pm.65}$	5.98$_{\pm.14}$
	+ TAIF-N	4.00$_{\pm.42}$	2.16$_{\pm.24}$	4.65$_{\pm.96}$	**17.32**$_{\pm.18}$	**4.47**$_{\pm.13}$	3.84$_{\pm.03}$	5.25$_{\pm.38}$	3.49$_{\pm.33}$	7.92$_{\pm1.35}$	**19.54**$_{\pm.32}$	**6.29**$_{\pm.13}$	**4.27**$_{\pm.04}$	6.62$_{\pm1.06}$	5.18$_{\pm.58}$	10.07$_{\pm1.48}$	**21.46**$_{\pm.43}$	**9.01**$_{\pm.17}$	**4.84**$_{\pm.10}$	8.08$_{\pm.66}$	8.53$_{\pm.70}$	**10.49**$_{\pm1.20}$	**22.09**$_{\pm.42}$	**12.88**$_{\pm.43}$	**5.34**$_{\pm.06}$
	Baseline	3.72$_{\pm.30}$	1.91$_{\pm.19}$	4.33$_{\pm.57}$	**17.41**$_{\pm.29}$	4.90$_{\pm.09}$	**3.85**$_{\pm.02}$	4.73$_{\pm.27}$	3.29$_{\pm.27}$	7.50$_{\pm.70}$	**19.85**$_{\pm.28}$	6.99$_{\pm.11}$	**4.44**$_{\pm.03}$	5.48$_{\pm.53}$	5.35$_{\pm.40}$	**9.66**$_{\pm1.06}$	**21.62**$_{\pm.23}$	9.78$_{\pm.22}$	**4.96**$_{\pm.01}$	7.20$_{\pm.52}$	8.36$_{\pm.26}$	**11.23**$_{\pm1.65}$	**22.78**$_{\pm.16}$	13.36$_{\pm.23}$	**5.41**$_{\pm.02}$
DiffPuter	+ TAIF-C	**3.58**$_{\pm.28}$	**1.65**$_{\pm.15}$	4.12$_{\pm1.13}$	17.83$_{\pm.28}$	5.18$_{\pm.12}$	3.85$_{\pm.03}$	4.35$_{\pm.51}$	**2.83**$_{\pm.18}$	7.45$_{\pm1.61}$	20.64$_{\pm.15}$	7.52$_{\pm.21}$	4.49$_{\pm.02}$	5.81$_{\pm.48}$	**4.68**$_{\pm.41}$	9.88$_{\pm1.70}$	22.81$_{\pm.27}$	10.59$_{\pm.29}$	5.14$_{\pm.03}$	7.07$_{\pm.47}$	**7.85**$_{\pm.48}$	12.04$_{\pm1.66}$	23.30$_{\pm.26}$	14.20$_{\pm.23}$	5.55$_{\pm.03}$
	+ TAIF-N	3.69$_{\pm.30}$	**1.66**$_{\pm.18}$	**3.75**$_{\pm1.05}$	**17.38**$_{\pm.27}$	8.57$_{\pm1.77}$	**3.61**$_{\pm.03}$	**4.04**$_{\pm.32}$	**2.54**$_{\pm.25}$	**6.40**$_{\pm1.74}$	19.92$_{\pm.19}$	12.38$_{\pm2.10}$	**4.16**$_{\pm.03}$	**5.31**$_{\pm.51}$	4.66$_{\pm.37}$	9.77$_{\pm1.74}$	21.64$_{\pm.29}$	14.11$_{\pm1.99}$	4.97$_{\pm.10}$	**6.65**$_{\pm.33}$	7.60$_{\pm.43}$	12.20$_{\pm1.72}$	22.89$_{\pm.34}$	21.31$_{\pm4.67}$	5.82$_{\pm.08}$
	Baseline	3.49$_{\pm.38}$	**1.49**$_{\pm.14}$	4.69$_{\pm1.44}$	**16.97**$_{\pm.18}$	**4.49**$_{\pm.16}$	**3.70**$_{\pm.03}$	4.34$_{\pm.44}$	2.75$_{\pm.26}$	7.63$_{\pm1.06}$	19.13$_{\pm.17}$	**6.28**$_{\pm.14}$	**4.17**$_{\pm.03}$	5.44$_{\pm.35}$	4.55$_{\pm.26}$	9.59$_{\pm1.07}$	**20.69**$_{\pm.18}$	**9.00**$_{\pm.22}$	4.73$_{\pm.02}$	**6.58**$_{\pm.43}$	7.38$_{\pm.40}$	11.18$_{\pm1.42}$	21.60$_{\pm.17}$	12.64$_{\pm.18}$	5.32$_{\pm.02}$
GRAPE	+ TAIF-C	**3.43**$_{\pm.27}$	1.54$_{\pm.13}$	4.56$_{\pm1.08}$	**16.83**$_{\pm.30}$	4.58$_{\pm.19}$	3.90$_{\pm.06}$	4.27$_{\pm.54}$	2.77$_{\pm.39}$	7.24$_{\pm1.11}$	**19.00**$_{\pm.19}$	6.36$_{\pm.23}$	4.31$_{\pm.04}$	**5.29**$_{\pm.42}$	**4.54**$_{\pm.35}$	10.04$_{\pm.84}$	20.70$_{\pm.20}$	9.04$_{\pm.21}$	4.82$_{\pm.03}$	6.63$_{\pm.50}$	7.50$_{\pm.46}$	11.30$_{\pm1.60}$	21.57$_{\pm.20}$	12.61$_{\pm.18}$	5.31$_{\pm.02}$
	+ TAIF-N	3.56$_{\pm.29}$	1.67$_{\pm.12}$	**4.30**$_{\pm1.03}$	17.02$_{\pm.18}$	4.54$_{\pm.15}$	3.74$_{\pm.03}$	**4.14**$_{\pm.48}$	**2.63**$_{\pm.19}$	**6.79**$_{\pm.97}$	19.11$_{\pm.16}$	6.31$_{\pm.16}$	4.18$_{\pm.03}$	5.61$_{\pm.53}$	4.61$_{\pm.38}$	**9.54**$_{\pm1.27}$	20.72$_{\pm.15}$	9.05$_{\pm.26}$	**4.72**$_{\pm.02}$	6.61$_{\pm.40}$	**7.30**$_{\pm.30}$	**10.77**$_{\pm1.11}$	**21.54**$_{\pm.13}$	**12.60**$_{\pm.15}$	**5.30**$_{\pm.02}$

3 Full Classification Experimental Results

This appendix provides complete experimental results across all methods, missing rates, and evaluation metrics in classification task.

Table S4: Downstream classification performance (Accuracy ↑) with XGBoost predictor across different missing rates. Best results in **bold**. TAIF-C: task signal concatenation, TAIF-N: neural refinement.

		MR = 30%						MR = 50%					
Base	Method	Dig.	Iris	Wine	Br.	Bean	Ges.	Dig.	Iris	Wine	Br.	Bean	Ges.
Mean	Baseline	**89.8**$_{\pm1.2}$	**92.2**$_{\pm3.3}$	**90.7**$_{\pm3.4}$	**94.9**$_{\pm1.7}$	90.7$_{\pm0.8}$	**55.9**$_{\pm1.9}$	**83.6**$_{\pm1.4}$	**86.7**$_{\pm4.4}$	**87.0**$_{\pm7.7}$	**94.9**$_{\pm1.6}$	**88.9**$_{\pm0.8}$	**50.4**$_{\pm1.5}$
	+ TAIF-C	**89.8**$_{\pm1.2}$	**92.2**$_{\pm3.3}$	**90.7**$_{\pm3.4}$	**94.9**$_{\pm1.7}$	90.7$_{\pm0.8}$	**56.1**$_{\pm1.7}$	**83.6**$_{\pm1.4}$	**86.7**$_{\pm4.4}$	**87.0**$_{\pm7.7}$	**94.9**$_{\pm1.6}$	**88.9**$_{\pm0.8}$	**50.4**$_{\pm1.5}$
	+ TAIF-N	89.0$_{\pm1.4}$	90.0$_{\pm5.3}$	89.5$_{\pm7.7}$	94.0$_{\pm1.7}$	88.6$_{\pm1.1}$	39.2$_{\pm2.4}$	73.2$_{\pm2.3}$	79.3$_{\pm5.7}$	80.2$_{\pm5.6}$	91.8$_{\pm1.4}$	85.6$_{\pm1.0}$	31.5$_{\pm1.0}$
KNN	Baseline	**94.3**$_{\pm1.0}$	87.4$_{\pm4.9}$	**96.0**$_{\pm4.2}$	**95.6**$_{\pm2.1}$	90.7$_{\pm0.7}$	59.8$_{\pm1.7}$	86.0$_{\pm1.4}$	**87.4**$_{\pm4.0}$	86.4$_{\pm4.5}$	95.0$_{\pm1.8}$	85.4$_{\pm1.3}$	50.4$_{\pm1.3}$
	+ TAIF-C	**94.3**$_{\pm1.0}$	**90.0**$_{\pm4.4}$	95.7$_{\pm3.1}$	95.5$_{\pm1.8}$	**91.5**$_{\pm0.8}$	**60.1**$_{\pm1.7}$	**88.2**$_{\pm1.5}$	85.6$_{\pm4.1}$	**89.2**$_{\pm3.8}$	**95.4**$_{\pm2.0}$	**89.7**$_{\pm0.9}$	**51.5**$_{\pm1.9}$
	+ TAIF-N	93.1$_{\pm1.2}$	88.5$_{\pm6.0}$	92.6$_{\pm5.2}$	95.2$_{\pm1.3}$	89.2$_{\pm1.0}$	39.7$_{\pm1.7}$	82.2$_{\pm1.8}$	84.1$_{\pm5.2}$	84.9$_{\pm3.4}$	94.2$_{\pm2.1}$	82.7$_{\pm1.2}$	32.3$_{\pm1.1}$
MICE	Baseline	**93.2**$_{\pm1.1}$	89.3$_{\pm6.0}$	**94.8**$_{\pm4.5}$	**97.2**$_{\pm0.9}$	**91.4**$_{\pm1.0}$	59.0$_{\pm1.6}$	87.2$_{\pm1.9}$	**85.6**$_{\pm4.4}$	86.4$_{\pm5.8}$	**95.4**$_{\pm1.8}$	**89.3**$_{\pm0.8}$	**52.0**$_{\pm1.9}$
	+ TAIF-C	92.9$_{\pm1.1}$	89.3$_{\pm4.9}$	94.4$_{\pm3.1}$	97.0$_{\pm1.5}$	**91.5**$_{\pm0.8}$	58.6$_{\pm1.9}$	**87.3**$_{\pm1.3}$	**85.6**$_{\pm5.3}$	**87.0**$_{\pm4.6}$	**96.4**$_{\pm1.0}$	**89.3**$_{\pm0.8}$	51.1$_{\pm2.1}$
	+ TAIF-N	91.5$_{\pm1.2}$	**90.7**$_{\pm4.3}$	93.5$_{\pm5.4}$	95.9$_{\pm1.1}$	89.5$_{\pm1.3}$	38.9$_{\pm1.8}$	82.4$_{\pm1.9}$	83.3$_{\pm2.9}$	85.2$_{\pm6.4}$	95.0$_{\pm1.4}$	86.9$_{\pm1.0}$	32.4$_{\pm2.3}$
MissForest	Baseline	93.3$_{\pm1.2}$	87.4$_{\pm8.0}$	93.2$_{\pm3.4}$	**95.8**$_{\pm1.4}$	90.8$_{\pm0.4}$	**60.8**$_{\pm1.7}$	86.4$_{\pm1.8}$	84.4$_{\pm6.2}$	84.6$_{\pm5.9}$	**95.3**$_{\pm1.8}$	89.1$_{\pm0.6}$	**54.3**$_{\pm1.4}$
	+ TAIF-C	**93.9**$_{\pm1.1}$	**90.0**$_{\pm5.0}$	**95.1**$_{\pm3.6}$	95.4$_{\pm1.3}$	**91.1**$_{\pm0.4}$	60.3$_{\pm1.0}$	**87.9**$_{\pm0.8}$	**86.3**$_{\pm5.1}$	**87.7**$_{\pm6.4}$	95.0$_{\pm1.6}$	**89.5**$_{\pm0.4}$	53.9$_{\pm1.7}$
	+ TAIF-N	91.8$_{\pm1.2}$	88.1$_{\pm4.1}$	92.9$_{\pm3.1}$	94.7$_{\pm2.3}$	89.2$_{\pm0.6}$	40.5$_{\pm2.1}$	83.2$_{\pm2.0}$	83.0$_{\pm5.1}$	87.0$_{\pm4.8}$	94.5$_{\pm2.2}$	86.1$_{\pm0.8}$	35.4$_{\pm1.5}$
GAIN	Baseline	91.0$_{\pm1.8}$	**86.7**$_{\pm5.8}$	89.5$_{\pm4.6}$	**95.6**$_{\pm1.6}$	89.4$_{\pm0.6}$	**53.5**$_{\pm1.8}$	72.0$_{\pm6.2}$	75.9$_{\pm10.2}$	71.6$_{\pm8.5}$	91.7$_{\pm4.2}$	**80.8**$_{\pm1.1}$	**44.9**$_{\pm3.0}$
	+ TAIF-C	90.6$_{\pm1.1}$	85.6$_{\pm6.2}$	87.0$_{\pm6.2}$	95.3$_{\pm1.6}$	89.6$_{\pm0.7}$	52.2$_{\pm2.1}$	69.6$_{\pm3.9}$	77.8$_{\pm7.6}$	72.5$_{\pm4.9}$	89.5$_{\pm4.7}$	76.1$_{\pm5.4}$	44.7$_{\pm2.2}$
	+ TAIF-N	**91.3**$_{\pm1.2}$	**86.7**$_{\pm5.8}$	**90.4**$_{\pm3.4}$	94.7$_{\pm1.3}$	88.7$_{\pm0.7}$	39.0$_{\pm2.1}$	**76.5**$_{\pm2.8}$	**78.5**$_{\pm8.4}$	**75.6**$_{\pm4.6}$	**93.4**$_{\pm2.0}$	76.4$_{\pm2.5}$	30.7$_{\pm1.2}$
MIWAE	Baseline	84.4$_{\pm1.5}$	81.1$_{\pm7.1}$	79.6$_{\pm4.4}$	**91.6**$_{\pm2.1}$	89.0$_{\pm0.6}$	**50.5**$_{\pm1.8}$	**69.8**$_{\pm1.6}$	71.9$_{\pm5.6}$	**70.1**$_{\pm5.2}$	**82.8**$_{\pm5.2}$	84.5$_{\pm0.5}$	**41.0**$_{\pm1.8}$
	+ TAIF-C	80.2$_{\pm2.1}$	84.8$_{\pm5.0}$	**84.0**$_{\pm6.5}$	90.4$_{\pm3.7}$	89.2$_{\pm0.5}$	48.5$_{\pm1.6}$	60.6$_{\pm2.2}$	71.9$_{\pm4.4}$	66.0$_{\pm8.1}$	81.6$_{\pm4.6}$	84.3$_{\pm0.5}$	40.0$_{\pm1.2}$
	+ TAIF-N	**86.1**$_{\pm1.8}$	**87.8**$_{\pm4.7}$	82.1$_{\pm4.2}$	90.3$_{\pm2.8}$	87.6$_{\pm0.7}$	41.9$_{\pm2.5}$	67.6$_{\pm3.7}$	**75.9**$_{\pm4.9}$	66.0$_{\pm4.3}$	77.8$_{\pm5.7}$	82.6$_{\pm0.7}$	32.6$_{\pm1.8}$
SoftImpute	Baseline	86.3$_{\pm3.1}$	87.4$_{\pm8.8}$	90.1$_{\pm4.2}$	95.4$_{\pm0.9}$	89.9$_{\pm0.3}$	**54.2**$_{\pm2.5}$	**74.2**$_{\pm5.7}$	83.7$_{\pm5.4}$	79.0$_{\pm9.6}$	94.1$_{\pm2.7}$	86.4$_{\pm1.0}$	**47.4**$_{\pm1.8}$
	+ TAIF-C	**88.5**$_{\pm1.3}$	**89.3**$_{\pm6.0}$	90.7$_{\pm4.8}$	95.2$_{\pm1.8}$	**90.0**$_{\pm0.5}$	49.7$_{\pm1.9}$	60.8$_{\pm7.9}$	**85.6**$_{\pm4.7}$	**81.8**$_{\pm8.3}$	**94.8**$_{\pm2.0}$	**88.1**$_{\pm0.8}$	**48.3**$_{\pm1.7}$
	+ TAIF-N	87.2$_{\pm1.6}$	88.5$_{\pm6.5}$	**92.3**$_{\pm4.3}$	**95.7**$_{\pm1.1}$	88.5$_{\pm0.4}$	40.5$_{\pm2.7}$	70.7$_{\pm2.6}$	83.7$_{\pm6.5}$	76.2$_{\pm9.9}$	94.1$_{\pm2.6}$	84.4$_{\pm1.4}$	31.6$_{\pm1.9}$
DiffPuter	Baseline	90.6$_{\pm1.5}$	87.4$_{\pm5.4}$	91.7$_{\pm5.9}$	**96.5**$_{\pm1.2}$	91.2$_{\pm1.2}$	57.9$_{\pm1.2}$	82.3$_{\pm1.6}$	79.6$_{\pm5.8}$	84.6$_{\pm4.0}$	94.4$_{\pm1.8}$	88.4$_{\pm1.2}$	49.8$_{\pm0.6}$
	+ TAIF-C	93.4$_{\pm1.2}$	90.4$_{\pm4.0}$	95.7$_{\pm3.0}$	95.9$_{\pm1.3}$	**91.9**$_{\pm1.1}$	**58.2**$_{\pm1.1}$	**86.6**$_{\pm1.5}$	87.0$_{\pm2.5}$	87.0$_{\pm6.4}$	**96.1**$_{\pm1.6}$	**89.8**$_{\pm1.0}$	**51.4**$_{\pm0.8}$
	+ TAIF-N	**93.5**$_{\pm0.8}$	**90.7**$_{\pm3.8}$	**95.4**$_{\pm3.5}$	96.4$_{\pm1.4}$	91.3$_{\pm1.1}$	58.1$_{\pm1.7}$	86.3$_{\pm1.5}$	**87.4**$_{\pm4.0}$	**89.5**$_{\pm7.4}$	**96.4**$_{\pm1.3}$	88.8$_{\pm1.3}$	50.4$_{\pm1.1}$

		MR = 70%						MR = 90%					
Base	Method	Dig.	Iris	Wine	Br.	Bean	Ges.	Dig.	Iris	Wine	Br.	Bean	Ges.
Mean	Baseline	**68.7**$_{\pm2.0}$	**65.6**$_{\pm6.7}$	**73.5**$_{\pm9.6}$	**92.8**$_{\pm1.7}$	**82.7**$_{\pm1.5}$	**46.0**$_{\pm1.8}$	**34.3**$_{\pm1.6}$	40.4$_{\pm5.1}$	**49.1**$_{\pm11.6}$	**83.1**$_{\pm2.3}$	**57.1**$_{\pm1.4}$	**39.8**$_{\pm1.3}$
	+ TAIF-C	**68.7**$_{\pm2.0}$	**65.6**$_{\pm6.7}$	**73.5**$_{\pm9.6}$	**92.8**$_{\pm1.7}$	**82.7**$_{\pm1.5}$	**46.0**$_{\pm1.8}$	**34.3**$_{\pm1.6}$	40.4$_{\pm5.1}$	**49.1**$_{\pm11.6}$	**83.1**$_{\pm2.3}$	**57.1**$_{\pm1.4}$	**39.8**$_{\pm1.3}$
	+ TAIF-N	48.7$_{\pm4.2}$	64.8$_{\pm7.3}$	63.0$_{\pm11.3}$	83.2$_{\pm4.0}$	78.6$_{\pm1.1}$	29.5$_{\pm1.7}$	19.0$_{\pm1.8}$	**44.8**$_{\pm3.8}$	45.7$_{\pm13.2}$	71.9$_{\pm4.1}$	55.9$_{\pm1.1}$	28.3$_{\pm0.9}$
KNN	Baseline	56.9$_{\pm1.7}$	66.3$_{\pm6.5}$	71.6$_{\pm6.0}$	90.7$_{\pm2.7}$	76.1$_{\pm0.8}$	42.5$_{\pm1.9}$	24.8$_{\pm2.7}$	44.1$_{\pm4.6}$	42.0$_{\pm14.0}$	78.5$_{\pm3.3}$	54.5$_{\pm1.5}$	38.0$_{\pm1.0}$
	+ TAIF-C	**71.8**$_{\pm1.2}$	**69.6**$_{\pm7.0}$	**72.8**$_{\pm7.3}$	**93.6**$_{\pm1.2}$	**83.0**$_{\pm1.3}$	**45.0**$_{\pm1.3}$	**31.3**$_{\pm2.5}$	**45.6**$_{\pm4.1}$	**50.3**$_{\pm9.0}$	**83.1**$_{\pm3.4}$	**56.0**$_{\pm1.5}$	**40.4**$_{\pm1.2}$
	+ TAIF-N	48.8$_{\pm3.7}$	63.7$_{\pm5.4}$	67.9$_{\pm6.8}$	89.2$_{\pm2.0}$	75.4$_{\pm0.9}$	31.5$_{\pm1.6}$	19.4$_{\pm3.5}$	45.2$_{\pm7.5}$	46.3$_{\pm10.0}$	78.6$_{\pm2.9}$	52.8$_{\pm1.1}$	30.8$_{\pm1.8}$
MICE	Baseline	67.2$_{\pm1.6}$	68.5$_{\pm5.3}$	**71.0**$_{\pm8.5}$	**93.9**$_{\pm1.2}$	81.7$_{\pm1.4}$	**45.2**$_{\pm1.4}$	**30.0**$_{\pm4.0}$	**46.3**$_{\pm3.9}$	42.0$_{\pm9.2}$	**80.2**$_{\pm4.0}$	**57.5**$_{\pm1.3}$	**39.5**$_{\pm1.3}$
	+ TAIF-C	**68.6**$_{\pm0.9}$	**68.9**$_{\pm5.3}$	68.2$_{\pm6.8}$	**93.7**$_{\pm1.6}$	81.8$_{\pm1.3}$	44.8$_{\pm1.7}$	29.7$_{\pm3.4}$	44.8$_{\pm4.7}$	**49.7**$_{\pm6.7}$	**82.2**$_{\pm3.9}$	55.4$_{\pm1.6}$	39.4$_{\pm1.0}$
	+ TAIF-N	60.2$_{\pm3.0}$	65.9$_{\pm6.0}$	64.8$_{\pm7.6}$	91.0$_{\pm1.4}$	79.2$_{\pm1.3}$	29.9$_{\pm2.1}$	22.2$_{\pm2.1}$	42.2$_{\pm3.7}$	46.6$_{\pm11.3}$	77.4$_{\pm4.4}$	55.9$_{\pm1.1}$	29.0$_{\pm1.3}$
MissForest	Baseline	68.7$_{\pm2.8}$	65.6$_{\pm7.8}$	67.3$_{\pm10.6}$	**93.4**$_{\pm1.6}$	82.4$_{\pm0.6}$	**46.8**$_{\pm1.2}$	25.1$_{\pm3.1}$	44.4$_{\pm7.6}$	46.3$_{\pm6.7}$	76.1$_{\pm4.9}$	54.0$_{\pm1.0}$	**38.6**$_{\pm1.2}$
	+ TAIF-C	**72.7**$_{\pm2.5}$	**68.9**$_{\pm7.3}$	**73.8**$_{\pm5.4}$	92.5$_{\pm1.8}$	82.7$_{\pm0.8}$	46.7$_{\pm1.6}$	**30.4**$_{\pm3.1}$	**45.6**$_{\pm4.7}$	**52.5**$_{\pm7.3}$	**83.2**$_{\pm2.9}$	**56.9**$_{\pm1.0}$	38.5$_{\pm1.1}$
	+ TAIF-N	60.8$_{\pm2.3}$	66.3$_{\pm7.2}$	66.7$_{\pm9.5}$	92.4$_{\pm1.8}$	79.7$_{\pm0.9}$	32.8$_{\pm1.0}$	21.6$_{\pm2.1}$	44.1$_{\pm3.6}$	45.7$_{\pm7.4}$	76.1$_{\pm4.2}$	54.3$_{\pm1.3}$	31.8$_{\pm1.3}$
GAIN	Baseline	37.5$_{\pm3.4}$	54.1$_{\pm8.9}$	50.9$_{\pm10.8}$	75.7$_{\pm5.3}$	47.3$_{\pm9.1}$	33.8$_{\pm2.7}$	14.4$_{\pm2.0}$	37.1$_{\pm6.5}$	34.4$_{\pm5.5}$	63.0$_{\pm4.8}$	**21.9**$_{\pm3.0}$	26.0$_{\pm3.5}$
	+ TAIF-C	36.5$_{\pm2.7}$	**61.5**$_{\pm11.7}$	44.4$_{\pm7.6}$	67.0$_{\pm6.8}$	**54.8**$_{\pm13.1}$	**37.7**$_{\pm4.1}$	13.7$_{\pm1.7}$	37.9$_{\pm4.0}$	30.9$_{\pm5.6}$	53.8$_{\pm12.9}$	15.9$_{\pm6.7}$	**30.8**$_{\pm2.3}$
	+ TAIF-N	**43.9**$_{\pm5.2}$	58.9$_{\pm7.3}$	**57.7**$_{\pm10.7}$	**79.1**$_{\pm4.8}$	50.0$_{\pm9.1}$	29.9$_{\pm1.2}$	**15.6**$_{\pm2.3}$	**38.7**$_{\pm6.2}$	**35.4**$_{\pm5.1}$	**65.9**$_{\pm3.8}$	21.6$_{\pm5.0}$	26.6$_{\pm1.7}$
MIWAE	Baseline	**43.4**$_{\pm1.0}$	56.3$_{\pm6.3}$	50.9$_{\pm11.5}$	66.4$_{\pm5.6}$	74.0$_{\pm0.7}$	**32.6**$_{\pm0.8}$	**17.1**$_{\pm2.7}$	38.1$_{\pm11.2}$	34.6$_{\pm5.6}$	53.1$_{\pm7.6}$	45.9$_{\pm0.8}$	**28.2**$_{\pm0.6}$
	+ TAIF-C	35.4$_{\pm1.7}$	51.9$_{\pm9.1}$	45.7$_{\pm11.5}$	**70.9**$_{\pm5.4}$	73.5$_{\pm0.6}$	31.7$_{\pm0.5}$	14.2$_{\pm1.7}$	38.1$_{\pm9.6}$	**37.0**$_{\pm6.8}$	**57.0**$_{\pm6.0}$	45.6$_{\pm0.8}$	28.3$_{\pm0.7}$
	+ TAIF-N	38.2$_{\pm3.4}$	**63.3**$_{\pm7.5}$	**51.2**$_{\pm7.9}$	59.8$_{\pm3.4}$	**75.4**$_{\pm1.1}$	30.4$_{\pm1.8}$	14.3$_{\pm1.4}$	**40.0**$_{\pm8.0}$	36.1$_{\pm8.6}$	48.4$_{\pm5.4}$	**50.7**$_{\pm0.6}$	27.6$_{\pm1.7}$
SoftImpute	Baseline	**50.6**$_{\pm5.1}$	63.0$_{\pm9.8}$	57.4$_{\pm10.9}$	91.6$_{\pm2.0}$	76.9$_{\pm0.8}$	**44.0**$_{\pm1.2}$	**18.4**$_{\pm1.2}$	40.0$_{\pm8.2}$	**41.0**$_{\pm7.1}$	74.5$_{\pm4.8}$	42.0$_{\pm5.2}$	**34.4**$_{\pm1.7}$
	+ TAIF-C	30.2$_{\pm6.9}$	**64.4**$_{\pm8.3}$	**68.2**$_{\pm9.2}$	**93.2**$_{\pm2.8}$	76.2$_{\pm10.2}$	**45.6**$_{\pm1.5}$	12.5$_{\pm2.1}$	42.2$_{\pm6.7}$	40.1$_{\pm12.0}$	**76.9**$_{\pm9.5}$	41.2$_{\pm12.5}$	**39.9**$_{\pm1.3}$
	+ TAIF-N	42.7$_{\pm5.3}$	63.0$_{\pm6.8}$	55.9$_{\pm13.4}$	89.9$_{\pm2.3}$	76.7$_{\pm0.8}$	27.8$_{\pm1.3}$	14.5$_{\pm2.8}$	**43.3**$_{\pm7.1}$	39.2$_{\pm10.5}$	71.9$_{\pm5.7}$	**47.3**$_{\pm5.3}$	28.0$_{\pm1.7}$
DiffPuter	Baseline	63.5$_{\pm1.6}$	60.4$_{\pm6.0}$	65.1$_{\pm6.4}$	91.2$_{\pm2.6}$	81.1$_{\pm1.1}$	44.3$_{\pm1.2}$	24.2$_{\pm1.3}$	38.1$_{\pm7.6}$	47.5$_{\pm5.0}$	75.6$_{\pm4.7}$	54.0$_{\pm0.7}$	36.9$_{\pm1.4}$
	+ TAIF-C	68.3$_{\pm2.6}$	60.0$_{\pm7.2}$	72.2$_{\pm6.4}$	**93.9**$_{\pm1.2}$	**83.7**$_{\pm1.2}$	**45.8**$_{\pm1.2}$	26.4$_{\pm2.3}$	43.0$_{\pm7.8}$	46.9$_{\pm5.9}$	**83.1**$_{\pm3.9}$	**54.7**$_{\pm0.7}$	37.4$_{\pm1.1}$
	+ TAIF-N	**69.7**$_{\pm2.0}$	**66.7**$_{\pm5.4}$	**72.2**$_{\pm5.2}$	**93.9**$_{\pm1.8}$	81.7$_{\pm1.2}$	44.7$_{\pm1.3}$	**30.6**$_{\pm2.6}$	**44.1**$_{\pm8.3}$	**49.4**$_{\pm7.4}$	82.6$_{\pm3.9}$	54.5$_{\pm0.8}$	**37.4**$_{\pm1.3}$

Table S5: Downstream classification performance (F1 Score ↑) with XGBoost predictor across different missing rates. Best results in **bold**. TAIF-C: task signal concatenation, TAIF-N: neural refinement.

		MR = 30%						MR = 50%					
Base	Method	Dig.	Iris	Wine	Br.	Bean	Ges.	Dig.	Iris	Wine	Br.	Bean	Ges.
Mean	Baseline	**89.9**$_{\pm1.2}$	**92.2**$_{\pm3.4}$	**90.8**$_{\pm3.4}$	**94.9**$_{\pm1.7}$	90.7$_{\pm0.8}$	**54.0**$_{\pm1.8}$	**83.6**$_{\pm1.5}$	**86.6**$_{\pm4.6}$	**87.1**$_{\pm7.7}$	**94.9**$_{\pm1.6}$	88.9$_{\pm0.8}$	**47.7**$_{\pm1.4}$
	+ TAIF-C	**89.9**$_{\pm1.2}$	**92.2**$_{\pm3.4}$	**90.8**$_{\pm3.4}$	**94.9**$_{\pm1.7}$	90.7$_{\pm0.8}$	**54.2**$_{\pm1.7}$	**83.6**$_{\pm1.5}$	**86.6**$_{\pm4.6}$	**87.1**$_{\pm7.7}$	**94.9**$_{\pm1.6}$	88.9$_{\pm0.8}$	**47.7**$_{\pm1.4}$
	+ TAIF-N	89.0$_{\pm1.4}$	89.9$_{\pm5.3}$	89.5$_{\pm7.7}$	93.9$_{\pm1.7}$	88.6$_{\pm1.1}$	38.8$_{\pm2.3}$	73.4$_{\pm2.2}$	78.9$_{\pm5.8}$	80.3$_{\pm5.4}$	91.8$_{\pm1.4}$	85.6$_{\pm1.0}$	31.8$_{\pm1.0}$
KNN	Baseline	**94.3**$_{\pm1.0}$	87.2$_{\pm5.1}$	**96.0**$_{\pm4.2}$	**95.6**$_{\pm2.1}$	90.8$_{\pm0.7}$	58.5$_{\pm1.7}$	86.0$_{\pm1.4}$	**87.3**$_{\pm4.1}$	86.3$_{\pm4.6}$	95.0$_{\pm1.7}$	86.3$_{\pm1.5}$	47.9$_{\pm1.2}$
	+ TAIF-C	**94.3**$_{\pm1.0}$	**89.9**$_{\pm4.5}$	95.7$_{\pm3.1}$	95.5$_{\pm1.8}$	**91.5**$_{\pm0.8}$	**58.7**$_{\pm1.6}$	**88.1**$_{\pm1.5}$	85.4$_{\pm4.3}$	**89.2**$_{\pm3.8}$	**95.4**$_{\pm2.0}$	**89.7**$_{\pm0.8}$	**49.0**$_{\pm1.9}$
	+ TAIF-N	93.1$_{\pm1.2}$	88.2$_{\pm6.2}$	92.6$_{\pm5.2}$	95.2$_{\pm1.3}$	89.3$_{\pm1.0}$	39.3$_{\pm1.6}$	82.2$_{\pm1.8}$	83.8$_{\pm5.5}$	84.7$_{\pm3.5}$	94.2$_{\pm2.1}$	83.0$_{\pm1.0}$	32.6$_{\pm0.9}$
MICE	Baseline	**93.2**$_{\pm1.1}$	89.2$_{\pm6.1}$	**94.7**$_{\pm4.5}$	**97.2**$_{\pm0.9}$	91.4$_{\pm1.0}$	57.6$_{\pm1.5}$	87.2$_{\pm1.9}$	**85.4**$_{\pm4.5}$	86.2$_{\pm5.9}$	95.8$_{\pm1.0}$	89.3$_{\pm0.8}$	**49.6**$_{\pm1.9}$
	+ TAIF-C	92.9$_{\pm1.1}$	89.1$_{\pm5.0}$	94.4$_{\pm3.1}$	97.0$_{\pm1.5}$	**91.5**$_{\pm0.8}$	57.1$_{\pm1.7}$	**87.3**$_{\pm1.4}$	85.4$_{\pm5.6}$	**87.0**$_{\pm4.6}$	**96.3**$_{\pm1.5}$	89.3$_{\pm0.8}$	48.5$_{\pm2.0}$
	+ TAIF-N	91.5$_{\pm1.2}$	**90.6**$_{\pm4.4}$	93.5$_{\pm5.4}$	95.9$_{\pm1.1}$	89.5$_{\pm1.3}$	38.6$_{\pm1.6}$	82.5$_{\pm1.9}$	83.1$_{\pm3.0}$	85.2$_{\pm6.3}$	95.0$_{\pm1.4}$	87.0$_{\pm1.0}$	32.8$_{\pm2.0}$
MissForest	Baseline	93.3$_{\pm1.2}$	87.2$_{\pm8.3}$	93.2$_{\pm3.5}$	**95.8**$_{\pm1.4}$	90.8$_{\pm0.4}$	**59.5**$_{\pm1.6}$	86.4$_{\pm1.8}$	84.1$_{\pm6.9}$	84.5$_{\pm5.9}$	**95.3**$_{\pm1.8}$	89.1$_{\pm0.6}$	**52.2**$_{\pm1.3}$
	+ TAIF-C	**93.9**$_{\pm1.1}$	**89.9**$_{\pm5.1}$	**95.1**$_{\pm3.6}$	95.4$_{\pm1.3}$	91.1$_{\pm0.4}$	58.9$_{\pm0.9}$	**87.9**$_{\pm0.9}$	**86.2**$_{\pm5.4}$	**87.6**$_{\pm6.4}$	95.0$_{\pm1.6}$	89.5$_{\pm0.4}$	51.7$_{\pm1.5}$
	+ TAIF-N	91.8$_{\pm1.2}$	88.0$_{\pm4.2}$	92.9$_{\pm3.1}$	94.7$_{\pm2.3}$	89.2$_{\pm0.6}$	40.2$_{\pm2.0}$	83.3$_{\pm1.9}$	82.6$_{\pm5.5}$	86.9$_{\pm4.8}$	94.5$_{\pm2.2}$	86.2$_{\pm0.7}$	35.2$_{\pm1.4}$
GAIN	Baseline	90.9$_{\pm1.8}$	86.3$_{\pm6.3}$	89.4$_{\pm4.6}$	**95.6**$_{\pm1.6}$	89.4$_{\pm0.6}$	**51.4**$_{\pm1.7}$	71.5$_{\pm6.4}$	74.6$_{\pm11.3}$	70.0$_{\pm10.2}$	91.7$_{\pm4.1}$	81.1$_{\pm1.1}$	**41.9**$_{\pm3.4}$
	+ TAIF-C	90.6$_{\pm1.1}$	85.5$_{\pm6.3}$	86.8$_{\pm6.6}$	95.3$_{\pm1.6}$	89.6$_{\pm0.7}$	50.4$_{\pm1.9}$	69.4$_{\pm4.1}$	77.4$_{\pm8.2}$	71.4$_{\pm4.9}$	89.1$_{\pm5.0}$	76.1$_{\pm5.4}$	40.7$_{\pm2.2}$
	+ TAIF-N	**91.3**$_{\pm1.2}$	**86.4**$_{\pm6.0}$	**90.4**$_{\pm3.5}$	94.7$_{\pm1.3}$	88.7$_{\pm0.7}$	38.9$_{\pm2.0}$	**76.5**$_{\pm2.8}$	**78.0**$_{\pm9.0}$	**74.5**$_{\pm5.6}$	**93.3**$_{\pm1.9}$	75.9$_{\pm2.9}$	30.7$_{\pm1.3}$
MIWAE	Baseline	84.2$_{\pm1.5}$	80.7$_{\pm7.5}$	79.7$_{\pm4.3}$	**91.7**$_{\pm2.1}$	89.0$_{\pm0.6}$	**48.1**$_{\pm1.7}$	**69.5**$_{\pm1.7}$	71.4$_{\pm6.3}$	**69.8**$_{\pm5.6}$	**83.1**$_{\pm5.1}$	84.5$_{\pm0.6}$	**36.8**$_{\pm1.9}$
	+ TAIF-C	80.0$_{\pm2.1}$	84.7$_{\pm5.1}$	**83.7**$_{\pm6.6}$	90.4$_{\pm3.6}$	89.2$_{\pm0.5}$	46.0$_{\pm1.5}$	60.2$_{\pm2.5}$	71.8$_{\pm4.7}$	65.3$_{\pm8.3}$	81.8$_{\pm4.4}$	84.3$_{\pm0.5}$	35.8$_{\pm1.4}$
	+ TAIF-N	**86.1**$_{\pm1.9}$	**87.6**$_{\pm4.8}$	82.0$_{\pm4.1}$	90.3$_{\pm2.7}$	87.6$_{\pm0.7}$	41.4$_{\pm2.2}$	67.6$_{\pm3.7}$	**75.5**$_{\pm4.9}$	65.2$_{\pm5.0}$	78.0$_{\pm5.8}$	82.7$_{\pm0.7}$	32.9$_{\pm1.6}$
SoftImpute	Baseline	86.2$_{\pm3.1}$	87.1$_{\pm9.3}$	90.1$_{\pm4.3}$	95.4$_{\pm0.9}$	89.9$_{\pm0.3}$	**52.5**$_{\pm2.0}$	**73.7**$_{\pm6.3}$	83.8$_{\pm5.3}$	78.8$_{\pm9.9}$	94.1$_{\pm2.6}$	86.4$_{\pm1.1}$	**44.7**$_{\pm1.1}$
	+ TAIF-C	**88.5**$_{\pm1.3}$	**89.1**$_{\pm6.1}$	90.8$_{\pm4.7}$	95.2$_{\pm1.8}$	90.0$_{\pm0.5}$	47.7$_{\pm1.9}$	60.5$_{\pm7.9}$	**85.4**$_{\pm4.8}$	**81.3**$_{\pm8.8}$	**94.8**$_{\pm2.0}$	88.1$_{\pm0.8}$	**44.7**$_{\pm1.7}$
	+ TAIF-N	87.2$_{\pm1.5}$	88.3$_{\pm6.6}$	**92.3**$_{\pm4.3}$	**95.7**$_{\pm1.1}$	88.5$_{\pm0.4}$	40.4$_{\pm2.5}$	70.7$_{\pm2.9}$	83.3$_{\pm6.7}$	75.8$_{\pm10.7}$	94.0$_{\pm2.6}$	84.4$_{\pm1.5}$	32.3$_{\pm1.8}$
DiffPuter	Baseline	90.6$_{\pm1.5}$	87.3$_{\pm5.6}$	91.7$_{\pm5.9}$	**96.5**$_{\pm1.2}$	91.2$_{\pm1.2}$	57.9$_{\pm1.2}$	82.2$_{\pm1.5}$	79.5$_{\pm5.8}$	84.5$_{\pm3.9}$	94.5$_{\pm1.8}$	88.4$_{\pm1.2}$	49.7$_{\pm0.7}$
	+ TAIF-C	93.3$_{\pm1.2}$	90.4$_{\pm4.0}$	95.7$_{\pm3.0}$	95.9$_{\pm1.3}$	**91.8**$_{\pm1.1}$	**58.2**$_{\pm1.1}$	**86.5**$_{\pm1.5}$	86.9$_{\pm2.5}$	87.0$_{\pm6.4}$	96.1$_{\pm1.6}$	**89.8**$_{\pm1.0}$	**51.4**$_{\pm0.7}$
	+ TAIF-N	**93.5**$_{\pm0.8}$	90.7$_{\pm3.8}$	95.3$_{\pm3.5}$	96.4$_{\pm1.4}$	91.3$_{\pm1.1}$	58.1$_{\pm1.6}$	86.3$_{\pm1.6}$	86.1$_{\pm4.4}$	89.5$_{\pm7.4}$	**96.4**$_{\pm1.3}$	88.8$_{\pm1.3}$	50.3$_{\pm1.1}$

		MR = 70%						MR = 90%					
Base	Method	Dig.	Iris	Wine	Br.	Bean	Ges.	Dig.	Iris	Wine	Br.	Bean	Ges.
Mean	Baseline	**68.7**$_{\pm2.1}$	**64.9**$_{\pm6.7}$	**73.3**$_{\pm9.7}$	**92.8**$_{\pm1.7}$	82.8$_{\pm1.5}$	**42.1**$_{\pm1.9}$	**33.8**$_{\pm1.5}$	34.0$_{\pm7.8}$	**47.1**$_{\pm12.0}$	**82.5**$_{\pm2.4}$	56.5$_{\pm1.6}$	**33.5**$_{\pm1.2}$
	+ TAIF-C	**68.7**$_{\pm2.1}$	**64.9**$_{\pm6.7}$	**73.3**$_{\pm9.7}$	**92.8**$_{\pm1.7}$	82.8$_{\pm1.5}$	**42.1**$_{\pm1.9}$	**33.8**$_{\pm1.5}$	34.0$_{\pm7.8}$	**47.1**$_{\pm12.0}$	**82.5**$_{\pm2.4}$	56.5$_{\pm1.6}$	**33.6**$_{\pm1.2}$
	+ TAIF-N	49.4$_{\pm4.1}$	64.0$_{\pm7.1}$	62.7$_{\pm11.7}$	83.0$_{\pm4.1}$	78.7$_{\pm1.1}$	28.5$_{\pm1.8}$	18.3$_{\pm1.9}$	**40.5**$_{\pm5.2}$	44.2$_{\pm13.5}$	71.2$_{\pm4.1}$	55.2$_{\pm1.1}$	27.5$_{\pm0.9}$
KNN	Baseline	57.0$_{\pm1.6}$	65.7$_{\pm6.6}$	71.5$_{\pm6.0}$	90.7$_{\pm2.8}$	77.3$_{\pm1.7}$	38.1$_{\pm1.6}$	24.7$_{\pm2.7}$	38.6$_{\pm6.8}$	39.9$_{\pm14.0}$	78.1$_{\pm3.4}$	54.4$_{\pm1.7}$	**32.8**$_{\pm1.0}$
	+ TAIF-C	**71.4**$_{\pm1.0}$	**69.2**$_{\pm7.3}$	**72.7**$_{\pm7.4}$	**93.6**$_{\pm1.1}$	**83.1**$_{\pm1.2}$	**40.2**$_{\pm1.3}$	**29.7**$_{\pm2.3}$	**41.1**$_{\pm6.2}$	**48.6**$_{\pm9.7}$	**82.6**$_{\pm3.6}$	55.4$_{\pm1.5}$	32.5$_{\pm1.2}$
	+ TAIF-N	49.7$_{\pm3.5}$	62.8$_{\pm4.7}$	67.8$_{\pm6.8}$	89.1$_{\pm2.0}$	75.9$_{\pm1.1}$	31.2$_{\pm1.5}$	19.1$_{\pm3.7}$	40.2$_{\pm9.7}$	44.9$_{\pm10.7}$	78.2$_{\pm2.9}$	53.0$_{\pm1.5}$	30.0$_{\pm1.8}$
MICE	Baseline	67.0$_{\pm1.5}$	68.2$_{\pm5.8}$	**70.9**$_{\pm8.7}$	92.5$_{\pm0.9}$	81.8$_{\pm1.3}$	**41.4**$_{\pm1.3}$	**29.6**$_{\pm4.0}$	**41.8**$_{\pm5.9}$	39.9$_{\pm10.0}$	80.0$_{\pm4.1}$	**56.8**$_{\pm1.3}$	**34.2**$_{\pm1.2}$
	+ TAIF-C	**68.2**$_{\pm0.8}$	**68.6**$_{\pm5.3}$	68.4$_{\pm7.0}$	**93.7**$_{\pm1.8}$	81.9$_{\pm1.3}$	40.4$_{\pm1.7}$	28.5$_{\pm3.0}$	40.0$_{\pm6.7}$	**48.1**$_{\pm6.6}$	**81.7**$_{\pm3.9}$	54.9$_{\pm1.6}$	33.0$_{\pm1.2}$
	+ TAIF-N	60.5$_{\pm3.0}$	65.5$_{\pm5.7}$	64.4$_{\pm8.3}$	90.9$_{\pm1.4}$	79.3$_{\pm1.3}$	29.5$_{\pm1.6}$	22.0$_{\pm2.2}$	36.2$_{\pm7.3}$	45.3$_{\pm12.0}$	77.1$_{\pm4.3}$	55.2$_{\pm1.1}$	28.2$_{\pm1.3}$
MissForest	Baseline	68.7$_{\pm2.8}$	64.9$_{\pm7.9}$	67.0$_{\pm10.5}$	**93.4**$_{\pm1.6}$	82.4$_{\pm0.7}$	**43.7**$_{\pm1.1}$	24.8$_{\pm3.1}$	39.1$_{\pm9.3}$	44.9$_{\pm7.7}$	75.6$_{\pm5.1}$	53.3$_{\pm1.0}$	**33.5**$_{\pm1.2}$
	+ TAIF-C	**72.6**$_{\pm2.3}$	**68.3**$_{\pm7.0}$	**73.7**$_{\pm5.4}$	92.5$_{\pm1.7}$	82.8$_{\pm0.7}$	43.7$_{\pm1.4}$	**29.6**$_{\pm3.0}$	**41.5**$_{\pm6.9}$	**51.0**$_{\pm8.2}$	**82.8**$_{\pm3.0}$	56.4$_{\pm1.0}$	33.5$_{\pm0.9}$
	+ TAIF-N	60.9$_{\pm2.2}$	65.7$_{\pm6.7}$	66.0$_{\pm10.3}$	92.4$_{\pm1.7}$	79.8$_{\pm0.9}$	32.4$_{\pm0.8}$	21.7$_{\pm2.1}$	39.1$_{\pm6.8}$	44.7$_{\pm7.3}$	75.7$_{\pm4.4}$	53.6$_{\pm1.3}$	30.9$_{\pm1.2}$
GAIN	Baseline	35.6$_{\pm3.6}$	50.9$_{\pm12.1}$	46.3$_{\pm13.4}$	72.8$_{\pm6.9}$	48.4$_{\pm8.4}$	27.4$_{\pm4.6}$	10.2$_{\pm2.1}$	26.8$_{\pm10.1}$	22.3$_{\pm7.4}$	51.8$_{\pm3.6}$	18.6$_{\pm3.8}$	18.5$_{\pm3.4}$
	+ TAIF-C	33.0$_{\pm3.6}$	**59.8**$_{\pm13.6}$	40.7$_{\pm8.2}$	60.8$_{\pm8.8}$	55.0$_{\pm11.4}$	**31.6**$_{\pm4.9}$	9.8$_{\pm1.5}$	26.4$_{\pm6.3}$	17.8$_{\pm5.5}$	41.6$_{\pm16.2}$	12.2$_{\pm5.3}$	**21.9**$_{\pm5.9}$
	+ TAIF-N	**43.2**$_{\pm5.7}$	57.8$_{\pm8.0}$	**55.4**$_{\pm12.8}$	**77.8**$_{\pm5.6}$	49.3$_{\pm10.1}$	28.4$_{\pm0.8}$	**12.0**$_{\pm3.5}$	**32.9**$_{\pm7.1}$	**26.9**$_{\pm6.6}$	**61.2**$_{\pm4.0}$	19.3$_{\pm6.2}$	21.5$_{\pm2.6}$
MIWAE	Baseline	**42.8**$_{\pm1.1}$	55.5$_{\pm7.3}$	47.9$_{\pm12.6}$	66.4$_{\pm5.9}$	74.1$_{\pm0.7}$	24.9$_{\pm1.4}$	**16.1**$_{\pm2.5}$	37.5$_{\pm11.4}$	28.7$_{\pm7.0}$	52.7$_{\pm8.4}$	45.5$_{\pm0.8}$	18.3$_{\pm1.0}$
	+ TAIF-C	34.1$_{\pm2.2}$	50.2$_{\pm8.9}$	41.5$_{\pm12.8}$	**71.2**$_{\pm5.3}$	73.6$_{\pm0.6}$	23.7$_{\pm1.1}$	12.0$_{\pm2.0}$	36.3$_{\pm8.8}$	**33.4**$_{\pm9.6}$	**56.3**$_{\pm5.7}$	45.3$_{\pm0.7}$	18.2$_{\pm1.2}$
	+ TAIF-N	38.2$_{\pm3.4}$	**62.5**$_{\pm7.3}$	**49.5**$_{\pm9.0}$	58.8$_{\pm4.2}$	75.5$_{\pm1.0}$	**29.6**$_{\pm1.7}$	13.6$_{\pm1.4}$	**38.7**$_{\pm8.5}$	32.6$_{\pm8.8}$	45.1$_{\pm8.7}$	50.2$_{\pm0.6}$	**26.5**$_{\pm1.6}$
SoftImpute	Baseline	**48.8**$_{\pm5.5}$	61.7$_{\pm11.0}$	54.5$_{\pm13.1}$	91.6$_{\pm2.0}$	76.8$_{\pm0.9}$	**40.0**$_{\pm1.1}$	**15.4**$_{\pm1.4}$	33.9$_{\pm10.4}$	**37.0**$_{\pm6.0}$	72.4$_{\pm6.1}$	40.3$_{\pm5.3}$	**30.5**$_{\pm1.4}$
	+ TAIF-C	27.6$_{\pm7.5}$	**63.6**$_{\pm9.3}$	**67.5**$_{\pm10.4}$	**93.2**$_{\pm2.8}$	75.8$_{\pm11.1}$	38.8$_{\pm1.7}$	10.1$_{\pm1.9}$	34.5$_{\pm12.5}$	35.6$_{\pm13.7}$	**74.7**$_{\pm13.0}$	41.3$_{\pm14.7}$	**31.6**$_{\pm1.3}$
	+ TAIF-N	41.8$_{\pm6.4}$	61.5$_{\pm6.5}$	54.8$_{\pm14.1}$	89.8$_{\pm2.3}$	76.7$_{\pm0.8}$	28.7$_{\pm1.3}$	13.6$_{\pm2.7}$	**37.7**$_{\pm10.9}$	35.1$_{\pm10.9}$	68.7$_{\pm9.0}$	46.3$_{\pm5.2}$	27.2$_{\pm1.6}$
DiffPuter	Baseline	63.3$_{\pm1.5}$	60.1$_{\pm5.8}$	64.5$_{\pm6.6}$	91.2$_{\pm2.6}$	81.2$_{\pm1.1}$	43.7$_{\pm1.2}$	23.8$_{\pm1.2}$	37.9$_{\pm7.5}$	47.2$_{\pm5.1}$	75.2$_{\pm4.9}$	54.1$_{\pm0.7}$	36.4$_{\pm1.5}$
	+ TAIF-C	67.8$_{\pm2.8}$	59.5$_{\pm7.1}$	72.0$_{\pm6.5}$	**93.9**$_{\pm1.1}$	**83.7**$_{\pm1.2}$	**45.4**$_{\pm1.2}$	25.1$_{\pm2.5}$	42.2$_{\pm7.4}$	45.9$_{\pm6.2}$	**82.9**$_{\pm3.8}$	54.4$_{\pm0.8}$	37.0$_{\pm1.0}$
	+ TAIF-N	**69.4**$_{\pm2.1}$	65.5$_{\pm6.6}$	72.1$_{\pm5.4}$	**93.9**$_{\pm1.7}$	81.8$_{\pm1.2}$	44.5$_{\pm1.3}$	**29.4**$_{\pm2.7}$	41.9$_{\pm9.2}$	47.9$_{\pm7.1}$	82.4$_{\pm3.8}$	**54.4**$_{\pm0.8}$	**36.8**$_{\pm1.3}$

Table S6: Imputation performance (MAE ↓) on normalized test set across different missing rates for classification datasets. Best results within each base method in **bold**.

Base	Method	MR = 30%						MR = 50%						MR = 70%						MR = 90%					
		Dig.	Iris	Wine	Br.	Bean	Ges.	Dig.	Iris	Wine	Br.	Bean	Ges.	Dig.	Iris	Wine	Br.	Bean	Ges.	Dig.	Iris	Wine	Br.	Bean	Ges.
Mean	Baseline	$\mathbf{0.20}_{\pm.00}$	$\mathbf{0.23}_{\pm.01}$	$\mathbf{0.19}_{\pm.01}$	$\mathbf{0.12}_{\pm.01}$	$\mathbf{0.11}_{\pm.00}$	$\mathbf{0.04}_{\pm.00}$	$\mathbf{0.20}_{\pm.00}$	$\mathbf{0.23}_{\pm.01}$	$\mathbf{0.19}_{\pm.01}$	$\mathbf{0.13}_{\pm.01}$	$\mathbf{0.11}_{\pm.00}$	$\mathbf{0.04}_{\pm.00}$	$\mathbf{0.20}_{\pm.00}$	$\mathbf{0.25}_{\pm.02}$	$\mathbf{0.22}_{\pm.01}$	$\mathbf{0.14}_{\pm.01}$	$\mathbf{0.12}_{\pm.00}$	$\mathbf{0.04}_{\pm.00}$	$\mathbf{0.20}_{\pm.00}$	$\mathbf{0.28}_{\pm.04}$	$\mathbf{0.26}_{\pm.01}$	$\mathbf{0.18}_{\pm.01}$	$\mathbf{0.13}_{\pm.00}$	$\mathbf{0.05}_{\pm.00}$
	+ TAIF-C	$\mathbf{0.20}_{\pm.00}$	$\mathbf{0.23}_{\pm.01}$	$\mathbf{0.19}_{\pm.01}$	$\mathbf{0.12}_{\pm.01}$	$\mathbf{0.11}_{\pm.00}$	$\mathbf{0.04}_{\pm.00}$	$\mathbf{0.20}_{\pm.00}$	$\mathbf{0.23}_{\pm.01}$	$\mathbf{0.19}_{\pm.01}$	$\mathbf{0.13}_{\pm.01}$	$\mathbf{0.11}_{\pm.00}$	$\mathbf{0.04}_{\pm.00}$	$\mathbf{0.20}_{\pm.00}$	$\mathbf{0.25}_{\pm.02}$	$\mathbf{0.22}_{\pm.01}$	$\mathbf{0.14}_{\pm.01}$	$\mathbf{0.12}_{\pm.00}$	$\mathbf{0.04}_{\pm.00}$	$\mathbf{0.20}_{\pm.00}$	$\mathbf{0.28}_{\pm.04}$	$\mathbf{0.26}_{\pm.01}$	$\mathbf{0.18}_{\pm.01}$	$\mathbf{0.13}_{\pm.00}$	$\mathbf{0.05}_{\pm.00}$
	+ TAIF-N	$0.16_{\pm.01}$	$0.30_{\pm.04}$	$0.16_{\pm.01}$	$0.09_{\pm.01}$	$0.06_{\pm.00}$	$0.06_{\pm.00}$	$0.18_{\pm.01}$	$0.47_{\pm.08}$	$0.18_{\pm.01}$	$0.10_{\pm.01}$	$0.06_{\pm.00}$	$0.06_{\pm.01}$	$0.20_{\pm.01}$	$1.37_{\pm.30}$	$0.24_{\pm.01}$	$0.12_{\pm.01}$	$0.07_{\pm.00}$	$0.07_{\pm.01}$	$0.24_{\pm.00}$	$2.66_{\pm.64}$	$0.51_{\pm.16}$	$0.20_{\pm.01}$	$0.15_{\pm.01}$	$0.08_{\pm.01}$
KNN	Baseline	$0.10_{\pm.00}$	$0.12_{\pm.02}$	$0.13_{\pm.01}$	$0.06_{\pm.00}$	$0.03_{\pm.00}$	$0.03_{\pm.00}$	$0.13_{\pm.00}$	$\mathbf{0.13}_{\pm.02}$	$0.16_{\pm.01}$	$0.08_{\pm.01}$	$0.07_{\pm.00}$	$\mathbf{0.03}_{\pm.00}$	$0.19_{\pm.00}$	$\mathbf{0.20}_{\pm.02}$	$0.20_{\pm.01}$	$0.12_{\pm.01}$	$0.09_{\pm.00}$	$\mathbf{0.04}_{\pm.00}$	$0.21_{\pm.00}$	$\mathbf{0.29}_{\pm.04}$	$0.29_{\pm.01}$	$0.18_{\pm.00}$	$0.10_{\pm.00}$	$\mathbf{0.06}_{\pm.00}$
	+ TAIF-C	$\mathbf{0.09}_{\pm.00}$	$\mathbf{0.12}_{\pm.03}$	$\mathbf{0.13}_{\pm.01}$	$\mathbf{0.06}_{\pm.00}$	$\mathbf{0.02}_{\pm.00}$	$\mathbf{0.03}_{\pm.00}$	$\mathbf{0.11}_{\pm.00}$	$0.14_{\pm.03}$	$\mathbf{0.14}_{\pm.01}$	$\mathbf{0.08}_{\pm.01}$	$\mathbf{0.05}_{\pm.00}$	$\mathbf{0.03}_{\pm.00}$	$\mathbf{0.15}_{\pm.00}$	$0.23_{\pm.02}$	$\mathbf{0.20}_{\pm.01}$	$\mathbf{0.11}_{\pm.01}$	$\mathbf{0.06}_{\pm.00}$	$0.05_{\pm.00}$	$\mathbf{0.20}_{\pm.00}$	$0.30_{\pm.05}$	$\mathbf{0.27}_{\pm.01}$	$\mathbf{0.17}_{\pm.01}$	$\mathbf{0.10}_{\pm.00}$	$\mathbf{0.06}_{\pm.00}$
	+ TAIF-N	$0.14_{\pm.01}$	$0.49_{\pm.07}$	$0.15_{\pm.01}$	$0.08_{\pm.01}$	$0.06_{\pm.00}$	$0.05_{\pm.00}$	$0.16_{\pm.01}$	$0.65_{\pm.07}$	$0.17_{\pm.01}$	$0.09_{\pm.01}$	$0.07_{\pm.00}$	$0.06_{\pm.00}$	$0.21_{\pm.01}$	$1.20_{\pm.40}$	$0.23_{\pm.01}$	$0.11_{\pm.01}$	$0.07_{\pm.00}$	$0.07_{\pm.01}$	$0.23_{\pm.01}$	$3.14_{\pm.84}$	$0.55_{\pm.08}$	$0.19_{\pm.01}$	$0.17_{\pm.02}$	$0.08_{\pm.01}$
MICE	Baseline	$\mathbf{0.12}_{\pm.00}$	$\mathbf{0.10}_{\pm.02}$	$0.17_{\pm.01}$	$\mathbf{0.04}_{\pm.00}$	$\mathbf{0.02}_{\pm.00}$	$\mathbf{0.02}_{\pm.00}$	$0.14_{\pm.00}$	$\mathbf{0.14}_{\pm.02}$	$0.16_{\pm.01}$	$\mathbf{0.06}_{\pm.00}$	$0.04_{\pm.00}$	$\mathbf{0.03}_{\pm.00}$	$0.18_{\pm.00}$	$\mathbf{0.22}_{\pm.03}$	$0.21_{\pm.01}$	$0.11_{\pm.01}$	$0.08_{\pm.01}$	$\mathbf{0.04}_{\pm.00}$	$\mathbf{0.21}_{\pm.00}$	$\mathbf{0.31}_{\pm.08}$	$0.31_{\pm.02}$	$0.19_{\pm.01}$	$0.12_{\pm.00}$	$\mathbf{0.06}_{\pm.00}$
	+ TAIF-C	$\mathbf{0.12}_{\pm.00}$	$0.11_{\pm.02}$	$\mathbf{0.17}_{\pm.01}$	$\mathbf{0.04}_{\pm.00}$	$\mathbf{0.02}_{\pm.00}$	$\mathbf{0.02}_{\pm.00}$	$\mathbf{0.14}_{\pm.00}$	$0.15_{\pm.03}$	$\mathbf{0.16}_{\pm.01}$	$\mathbf{0.06}_{\pm.00}$	$\mathbf{0.03}_{\pm.00}$	$\mathbf{0.03}_{\pm.00}$	$\mathbf{0.18}_{\pm.00}$	$0.23_{\pm.04}$	$\mathbf{0.21}_{\pm.01}$	$\mathbf{0.10}_{\pm.01}$	$\mathbf{0.08}_{\pm.01}$	$\mathbf{0.04}_{\pm.00}$	$\mathbf{0.21}_{\pm.00}$	$0.32_{\pm.05}$	$\mathbf{0.31}_{\pm.03}$	$\mathbf{0.18}_{\pm.01}$	$\mathbf{0.12}_{\pm.00}$	$\mathbf{0.06}_{\pm.00}$
	+ TAIF-N	$0.15_{\pm.02}$	$0.44_{\pm.08}$	$0.16_{\pm.01}$	$0.08_{\pm.01}$	$0.06_{\pm.00}$	$0.06_{\pm.00}$	$0.16_{\pm.01}$	$0.62_{\pm.13}$	$0.18_{\pm.01}$	$0.09_{\pm.01}$	$0.06_{\pm.00}$	$0.06_{\pm.00}$	$0.19_{\pm.01}$	$1.10_{\pm.24}$	$0.25_{\pm.02}$	$0.11_{\pm.00}$	$0.07_{\pm.00}$	$0.07_{\pm.01}$	$0.23_{\pm.00}$	$4.10_{\pm2.49}$	$0.48_{\pm.08}$	$0.18_{\pm.01}$	$0.17_{\pm.01}$	$0.08_{\pm.01}$
MissForest	Baseline	$0.08_{\pm.00}$	$0.12_{\pm.03}$	$0.13_{\pm.01}$	$\mathbf{0.04}_{\pm.00}$	$\mathbf{0.02}_{\pm.00}$	$\mathbf{0.02}_{\pm.00}$	$0.10_{\pm.00}$	$\mathbf{0.14}_{\pm.03}$	$0.16_{\pm.01}$	$\mathbf{0.06}_{\pm.01}$	$\mathbf{0.02}_{\pm.00}$	$\mathbf{0.02}_{\pm.00}$	$0.14_{\pm.00}$	$\mathbf{0.21}_{\pm.04}$	$0.22_{\pm.01}$	$\mathbf{0.09}_{\pm.00}$	$\mathbf{0.04}_{\pm.00}$	$\mathbf{0.03}_{\pm.00}$	$0.22_{\pm.00}$	$0.34_{\pm.05}$	$0.31_{\pm.02}$	$0.19_{\pm.01}$	$\mathbf{0.09}_{\pm.00}$	$\mathbf{0.06}_{\pm.00}$
	+ TAIF-C	$\mathbf{0.08}_{\pm.00}$	$\mathbf{0.11}_{\pm.03}$	$\mathbf{0.13}_{\pm.01}$	$0.05_{\pm.00}$	$\mathbf{0.02}_{\pm.00}$	$\mathbf{0.02}_{\pm.00}$	$\mathbf{0.10}_{\pm.00}$	$0.14_{\pm.03}$	$\mathbf{0.15}_{\pm.02}$	$\mathbf{0.06}_{\pm.01}$	$\mathbf{0.02}_{\pm.00}$	$\mathbf{0.02}_{\pm.00}$	$\mathbf{0.14}_{\pm.00}$	$0.23_{\pm.03}$	$\mathbf{0.21}_{\pm.01}$	$\mathbf{0.09}_{\pm.00}$	$\mathbf{0.04}_{\pm.00}$	$\mathbf{0.03}_{\pm.00}$	$\mathbf{0.22}_{\pm.00}$	$\mathbf{0.33}_{\pm.07}$	$\mathbf{0.29}_{\pm.02}$	$\mathbf{0.18}_{\pm.01}$	$\mathbf{0.09}_{\pm.00}$	$\mathbf{0.06}_{\pm.00}$
	+ TAIF-N	$0.14_{\pm.01}$	$0.50_{\pm.09}$	$0.16_{\pm.01}$	$0.08_{\pm.01}$	$0.06_{\pm.00}$	$0.06_{\pm.00}$	$0.15_{\pm.02}$	$0.62_{\pm.09}$	$0.18_{\pm.02}$	$0.08_{\pm.01}$	$0.06_{\pm.00}$	$0.06_{\pm.00}$	$0.18_{\pm.01}$	$1.16_{\pm.32}$	$0.24_{\pm.02}$	$0.10_{\pm.01}$	$0.07_{\pm.00}$	$0.07_{\pm.00}$	$0.23_{\pm.01}$	$2.73_{\pm.95}$	$0.57_{\pm.21}$	$0.19_{\pm.01}$	$0.15_{\pm.01}$	$0.09_{\pm.01}$
GAIN	Baseline	$\mathbf{0.13}_{\pm.01}$	$\mathbf{0.13}_{\pm.02}$	$0.19_{\pm.03}$	$0.06_{\pm.00}$	$0.05_{\pm.00}$	$0.04_{\pm.00}$	$0.23_{\pm.02}$	$0.18_{\pm.06}$	$0.27_{\pm.03}$	$0.12_{\pm.01}$	$0.11_{\pm.03}$	$0.18_{\pm.04}$	$0.28_{\pm.01}$	$0.27_{\pm.05}$	$0.37_{\pm.03}$	$0.23_{\pm.02}$	$0.26_{\pm.05}$	$0.26_{\pm.04}$	$0.29_{\pm.02}$	$0.36_{\pm.07}$	$0.37_{\pm.04}$	$0.28_{\pm.02}$	$0.25_{\pm.04}$	$0.16_{\pm.03}$
	+ TAIF-C	$\mathbf{0.13}_{\pm.00}$	$0.14_{\pm.03}$	$0.20_{\pm.02}$	$0.07_{\pm.00}$	$0.05_{\pm.00}$	$\mathbf{0.04}_{\pm.00}$	$0.23_{\pm.02}$	$\mathbf{0.18}_{\pm.02}$	$0.26_{\pm.02}$	$0.12_{\pm.02}$	$0.13_{\pm.03}$	$\mathbf{0.15}_{\pm.04}$	$0.28_{\pm.01}$	$\mathbf{0.24}_{\pm.03}$	$0.36_{\pm.03}$	$0.23_{\pm.01}$	$0.25_{\pm.03}$	$\mathbf{0.24}_{\pm.03}$	$0.31_{\pm.02}$	$\mathbf{0.36}_{\pm.11}$	$0.37_{\pm.04}$	$0.29_{\pm.02}$	$0.26_{\pm.04}$	$\mathbf{0.12}_{\pm.03}$
	+ TAIF-N	$0.15_{\pm.01}$	$0.38_{\pm.09}$	$\mathbf{0.17}_{\pm.01}$	$\mathbf{0.08}_{\pm.00}$	$\mathbf{0.06}_{\pm.00}$	$0.05_{\pm.00}$	$\mathbf{0.17}_{\pm.01}$	$0.59_{\pm.12}$	$\mathbf{0.20}_{\pm.02}$	$\mathbf{0.09}_{\pm.01}$	$\mathbf{0.08}_{\pm.01}$	$0.06_{\pm.01}$	$\mathbf{0.21}_{\pm.01}$	$0.92_{\pm.29}$	$\mathbf{0.27}_{\pm.02}$	$\mathbf{0.13}_{\pm.01}$	$\mathbf{0.12}_{\pm.02}$	$0.07_{\pm.01}$	$\mathbf{0.28}_{\pm.04}$	$1.94_{\pm.87}$	$\mathbf{0.53}_{\pm.13}$	$\mathbf{0.21}_{\pm.02}$	$\mathbf{0.21}_{\pm.02}$	$0.09_{\pm.02}$
MIWAE	Baseline	$0.27_{\pm.00}$	$0.33_{\pm.03}$	$0.49_{\pm.04}$	$0.33_{\pm.00}$	$0.12_{\pm.00}$	$0.05_{\pm.00}$	$0.27_{\pm.00}$	$0.34_{\pm.02}$	$0.50_{\pm.03}$	$0.34_{\pm.01}$	$0.12_{\pm.00}$	$0.05_{\pm.00}$	$0.27_{\pm.00}$	$0.37_{\pm.03}$	$0.53_{\pm.02}$	$0.35_{\pm.01}$	$0.12_{\pm.00}$	$0.06_{\pm.00}$	$0.27_{\pm.00}$	$0.48_{\pm.15}$	$0.58_{\pm.04}$	$0.38_{\pm.01}$	$0.13_{\pm.00}$	$0.07_{\pm.00}$
	+ TAIF-C	$0.42_{\pm.00}$	$\mathbf{0.28}_{\pm.04}$	$0.52_{\pm.03}$	$\mathbf{0.17}_{\pm.01}$	$0.12_{\pm.00}$	$0.06_{\pm.00}$	$0.42_{\pm.00}$	$\mathbf{0.28}_{\pm.03}$	$0.53_{\pm.03}$	$\mathbf{0.18}_{\pm.01}$	$0.12_{\pm.00}$	$0.06_{\pm.00}$	$0.42_{\pm.00}$	$\mathbf{0.31}_{\pm.04}$	$0.54_{\pm.02}$	$\mathbf{0.19}_{\pm.00}$	$0.12_{\pm.00}$	$0.06_{\pm.00}$	$0.42_{\pm.00}$	$\mathbf{0.36}_{\pm.07}$	$0.58_{\pm.02}$	$\mathbf{0.23}_{\pm.01}$	$0.13_{\pm.00}$	$0.07_{\pm.00}$
	+ TAIF-N	$\mathbf{0.16}_{\pm.01}$	$0.36_{\pm.05}$	$\mathbf{0.22}_{\pm.02}$	$0.10_{\pm.00}$	$\mathbf{0.06}_{\pm.00}$	$\mathbf{0.05}_{\pm.00}$	$\mathbf{0.19}_{\pm.01}$	$0.47_{\pm.06}$	$\mathbf{0.24}_{\pm.02}$	$0.13_{\pm.01}$	$\mathbf{0.07}_{\pm.00}$	$\mathbf{0.06}_{\pm.00}$	$\mathbf{0.22}_{\pm.01}$	$1.19_{\pm.33}$	$\mathbf{0.31}_{\pm.02}$	$0.18_{\pm.01}$	$\mathbf{0.07}_{\pm.00}$	$\mathbf{0.06}_{\pm.00}$	$\mathbf{0.25}_{\pm.00}$	$1.98_{\pm.30}$	$\mathbf{0.41}_{\pm.05}$	$0.28_{\pm.03}$	$\mathbf{0.15}_{\pm.01}$	$\mathbf{0.07}_{\pm.00}$
SoftImpute	Baseline	$0.18_{\pm.00}$	$0.22_{\pm.09}$	$0.17_{\pm.02}$	$0.08_{\pm.01}$	$0.08_{\pm.00}$	$0.04_{\pm.00}$	$0.19_{\pm.00}$	$0.24_{\pm.04}$	$0.20_{\pm.04}$	$0.09_{\pm.01}$	$0.09_{\pm.00}$	$0.04_{\pm.00}$	$0.19_{\pm.00}$	$0.35_{\pm.05}$	$0.33_{\pm.09}$	$0.12_{\pm.01}$	$0.11_{\pm.00}$	$0.05_{\pm.00}$	$0.25_{\pm.01}$	$0.49_{\pm.06}$	$0.45_{\pm.04}$	$0.22_{\pm.03}$	$0.26_{\pm.01}$	$0.11_{\pm.00}$
	+ TAIF-C	$0.20_{\pm.00}$	$\mathbf{0.17}_{\pm.03}$	$\mathbf{0.16}_{\pm.01}$	$\mathbf{0.08}_{\pm.01}$	$0.12_{\pm.00}$	$0.04_{\pm.00}$	$0.20_{\pm.00}$	$\mathbf{0.25}_{\pm.12}$	$\mathbf{0.17}_{\pm.01}$	$\mathbf{0.09}_{\pm.01}$	$0.13_{\pm.00}$	$0.04_{\pm.01}$	$0.20_{\pm.00}$	$\mathbf{0.37}_{\pm.07}$	$\mathbf{0.24}_{\pm.03}$	$\mathbf{0.11}_{\pm.01}$	$0.14_{\pm.01}$	$0.05_{\pm.01}$	$\mathbf{0.23}_{\pm.00}$	$\mathbf{0.45}_{\pm.09}$	$0.46_{\pm.05}$	$\mathbf{0.19}_{\pm.06}$	$0.22_{\pm.02}$	$0.16_{\pm.04}$
	+ TAIF-N	$\mathbf{0.17}_{\pm.01}$	$0.47_{\pm.11}$	$0.18_{\pm.01}$	$\mathbf{0.08}_{\pm.01}$	$\mathbf{0.06}_{\pm.00}$	$\mathbf{0.05}_{\pm.00}$	$\mathbf{0.18}_{\pm.00}$	$0.61_{\pm.10}$	$0.20_{\pm.01}$	$\mathbf{0.09}_{\pm.01}$	$\mathbf{0.06}_{\pm.00}$	$\mathbf{0.06}_{\pm.00}$	$\mathbf{0.22}_{\pm.01}$	$1.12_{\pm.26}$	$0.29_{\pm.03}$	$\mathbf{0.11}_{\pm.01}$	$\mathbf{0.08}_{\pm.00}$	$\mathbf{0.06}_{\pm.01}$	$0.25_{\pm.01}$	$2.52_{\pm.55}$	$0.59_{\pm.11}$	$0.21_{\pm.03}$	$\mathbf{0.20}_{\pm.02}$	$\mathbf{0.08}_{\pm.00}$
DiffPuter	Baseline	$0.16_{\pm.00}$	$0.18_{\pm.01}$	$0.16_{\pm.01}$	$0.08_{\pm.00}$	$0.07_{\pm.00}$	$\mathbf{0.03}_{\pm.00}$	$0.18_{\pm.00}$	$0.21_{\pm.01}$	$0.18_{\pm.01}$	$0.10_{\pm.01}$	$0.09_{\pm.00}$	$\mathbf{0.03}_{\pm.00}$	$0.19_{\pm.00}$	$0.25_{\pm.02}$	$0.22_{\pm.01}$	$0.13_{\pm.00}$	$0.10_{\pm.00}$	$\mathbf{0.04}_{\pm.00}$	$0.20_{\pm.00}$	$0.28_{\pm.04}$	$0.26_{\pm.01}$	$0.19_{\pm.00}$	$0.12_{\pm.00}$	$\mathbf{0.05}_{\pm.00}$
	+ TAIF-C	$\mathbf{0.15}_{\pm.00}$	$\mathbf{0.15}_{\pm.01}$	$\mathbf{0.14}_{\pm.01}$	$\mathbf{0.08}_{\pm.00}$	$\mathbf{0.06}_{\pm.00}$	$\mathbf{0.03}_{\pm.00}$	$\mathbf{0.17}_{\pm.00}$	$\mathbf{0.18}_{\pm.02}$	$\mathbf{0.16}_{\pm.01}$	$\mathbf{0.10}_{\pm.00}$	$\mathbf{0.08}_{\pm.00}$	$\mathbf{0.03}_{\pm.00}$	$\mathbf{0.19}_{\pm.00}$	$\mathbf{0.23}_{\pm.02}$	$\mathbf{0.21}_{\pm.01}$	$\mathbf{0.13}_{\pm.00}$	$\mathbf{0.09}_{\pm.00}$	$\mathbf{0.04}_{\pm.00}$	$\mathbf{0.20}_{\pm.00}$	$\mathbf{0.28}_{\pm.04}$	$\mathbf{0.26}_{\pm.01}$	$\mathbf{0.18}_{\pm.00}$	$\mathbf{0.12}_{\pm.00}$	$\mathbf{0.05}_{\pm.00}$
	+ TAIF-N	$0.16_{\pm.00}$	$0.16_{\pm.02}$	$\mathbf{0.14}_{\pm.01}$	$\mathbf{0.08}_{\pm.00}$	$0.07_{\pm.00}$	$\mathbf{0.03}_{\pm.00}$	$\mathbf{0.17}_{\pm.00}$	$\mathbf{0.17}_{\pm.02}$	$0.15_{\pm.01}$	$\mathbf{0.10}_{\pm.00}$	$0.09_{\pm.00}$	$\mathbf{0.03}_{\pm.00}$	$\mathbf{0.19}_{\pm.00}$	$\mathbf{0.23}_{\pm.02}$	$0.20_{\pm.01}$	$\mathbf{0.13}_{\pm.00}$	$0.10_{\pm.00}$	$\mathbf{0.04}_{\pm.00}$	$\mathbf{0.20}_{\pm.00}$	$\mathbf{0.28}_{\pm.04}$	$\mathbf{0.26}_{\pm.01}$	$\mathbf{0.18}_{\pm.00}$	$\mathbf{0.12}_{\pm.00}$	$\mathbf{0.05}_{\pm.00}$

Table S7: Downstream prediction performance (MAE ↓) on test set across different missing rates under **MAR** setting. Kin8nm values scaled by ×100. Best results within each base method in **bold**.

Base	Method	Missing Rate = 30%			Missing Rate = 50%			Missing Rate = 70%			Missing Rate = 90%		
		Hou.	Yac.	$\text{Kin.}_{\times 100}$	Hou.	Yac.	$\text{Kin.}_{\times 100}$	Hou.	Yac.	$\text{Kin.}_{\times 100}$	Hou.	Yac.	$\text{Kin.}_{\times 100}$
Mean	Baseline	$3.25_{\pm.52}$	$2.95_{\pm1.89}$	$\mathbf{15.28}_{\pm.54}$	$\mathbf{3.74}_{\pm.38}$	$\mathbf{4.67}_{\pm3.03}$	$16.73_{\pm.75}$	$\mathbf{4.40}_{\pm.39}$	$6.78_{\pm4.64}$	$18.09_{\pm.98}$	$4.69_{\pm.48}$	$\mathbf{7.62}_{\pm5.13}$	$\mathbf{19.25}_{\pm1.22}$
	+ TAIF-C	$\mathbf{3.25}_{\pm.52}$	$\mathbf{2.95}_{\pm1.89}$	$15.28_{\pm.54}$	$3.74_{\pm.38}$	$4.67_{\pm3.03}$	$16.73_{\pm.75}$	$4.40_{\pm.39}$	$6.78_{\pm4.64}$	$18.09_{\pm.98}$	$\mathbf{4.69}_{\pm.48}$	$7.62_{\pm5.13}$	$19.25_{\pm1.22}$
	+ TAIF-N	$3.45_{\pm.58}$	$3.50_{\pm2.22}$	$15.32_{\pm.49}$	$4.48_{\pm.49}$	$5.28_{\pm3.48}$	$\mathbf{16.70}_{\pm.70}$	$5.15_{\pm.55}$	$\mathbf{6.70}_{\pm4.55}$	$\mathbf{18.07}_{\pm.94}$	$5.42_{\pm.57}$	$7.66_{\pm5.02}$	$19.39_{\pm1.33}$
Median	Baseline	$\mathbf{3.24}_{\pm.53}$	$\mathbf{2.96}_{\pm1.96}$	$15.28_{\pm.55}$	$\mathbf{3.78}_{\pm.37}$	$\mathbf{4.77}_{\pm3.11}$	$16.72_{\pm.75}$	$\mathbf{4.38}_{\pm.42}$	$\mathbf{6.77}_{\pm4.61}$	$\mathbf{18.08}_{\pm.98}$	$\mathbf{4.64}_{\pm.44}$	$\mathbf{7.43}_{\pm5.04}$	$\mathbf{19.24}_{\pm1.21}$
	+ TAIF-C	$3.24_{\pm.53}$	$2.96_{\pm1.96}$	$15.28_{\pm.55}$	$3.78_{\pm.37}$	$4.77_{\pm3.11}$	$16.72_{\pm.75}$	$4.38_{\pm.42}$	$6.77_{\pm4.61}$	$18.08_{\pm.98}$	$4.64_{\pm.44}$	$7.43_{\pm5.04}$	$19.24_{\pm1.21}$
	+ TAIF-N	$3.48_{\pm.54}$	$3.38_{\pm2.09}$	$\mathbf{15.26}_{\pm.44}$	$4.39_{\pm.52}$	$4.87_{\pm3.07}$	$\mathbf{16.72}_{\pm.70}$	$5.24_{\pm.57}$	$6.83_{\pm4.74}$	$18.09_{\pm.95}$	$5.29_{\pm.40}$	$7.65_{\pm5.00}$	$19.47_{\pm1.28}$
KNN	Baseline	$3.13_{\pm.39}$	$3.01_{\pm1.97}$	$15.64_{\pm.62}$	$4.02_{\pm.32}$	$4.81_{\pm3.21}$	$17.06_{\pm.74}$	$4.52_{\pm.56}$	$\mathbf{6.65}_{\pm4.53}$	$18.30_{\pm1.10}$	$\mathbf{4.73}_{\pm.47}$	$\mathbf{7.67}_{\pm5.12}$	$19.49_{\pm1.42}$
	+ TAIF-C	$\mathbf{3.09}_{\pm.37}$	$\mathbf{2.98}_{\pm1.93}$	$15.51_{\pm.64}$	$\mathbf{3.58}_{\pm.32}$	$\mathbf{4.64}_{\pm3.02}$	$17.33_{\pm.81}$	$\mathbf{4.33}_{\pm.39}$	$6.67_{\pm4.59}$	$19.06_{\pm1.42}$	$5.04_{\pm.42}$	$7.76_{\pm5.21}$	$20.58_{\pm1.65}$
	+ TAIF-N	$3.34_{\pm.54}$	$3.13_{\pm1.94}$	$\mathbf{15.40}_{\pm.56}$	$4.30_{\pm.38}$	$5.40_{\pm3.57}$	$\mathbf{16.84}_{\pm.61}$	$4.92_{\pm.43}$	$7.15_{\pm4.96}$	$\mathbf{18.11}_{\pm1.03}$	$5.36_{\pm.45}$	$7.75_{\pm5.02}$	$\mathbf{19.37}_{\pm1.45}$
MICE	Baseline	$3.38_{\pm.45}$	$\mathbf{3.09}_{\pm1.88}$	$15.29_{\pm.55}$	$3.92_{\pm.43}$	$4.90_{\pm3.24}$	$16.86_{\pm.65}$	$4.67_{\pm.42}$	$\mathbf{6.52}_{\pm4.47}$	$18.25_{\pm.97}$	$5.04_{\pm.71}$	$7.71_{\pm5.17}$	$\mathbf{19.43}_{\pm1.29}$
	+ TAIF-C	$\mathbf{3.23}_{\pm.29}$	$3.11_{\pm1.95}$	$15.83_{\pm.44}$	$\mathbf{3.83}_{\pm.48}$	$\mathbf{4.71}_{\pm3.06}$	$17.66_{\pm.65}$	$\mathbf{4.60}_{\pm.37}$	$6.60_{\pm4.57}$	$19.29_{\pm.85}$	$\mathbf{4.94}_{\pm.33}$	$7.88_{\pm5.31}$	$20.65_{\pm1.45}$
	+ TAIF-N	$3.45_{\pm.53}$	$3.34_{\pm2.14}$	$\mathbf{15.29}_{\pm.50}$	$4.23_{\pm.55}$	$5.07_{\pm3.30}$	$\mathbf{16.78}_{\pm.57}$	$5.10_{\pm.64}$	$6.96_{\pm4.69}$	$\mathbf{18.17}_{\pm.92}$	$5.37_{\pm.45}$	$\mathbf{7.60}_{\pm4.98}$	$19.49_{\pm1.34}$
MissForest	Baseline	$3.11_{\pm.32}$	$\mathbf{2.98}_{\pm1.95}$	$15.50_{\pm.54}$	$3.90_{\pm.45}$	$\mathbf{4.69}_{\pm3.10}$	$17.16_{\pm.75}$	$4.74_{\pm.55}$	$6.81_{\pm4.56}$	$18.60_{\pm1.12}$	$\mathbf{4.90}_{\pm.62}$	$\mathbf{7.55}_{\pm5.09}$	$19.53_{\pm1.33}$
	+ TAIF-C	$\mathbf{2.97}_{\pm.36}$	$3.00_{\pm1.91}$	$\mathbf{15.40}_{\pm.44}$	$\mathbf{3.57}_{\pm.33}$	$4.76_{\pm3.10}$	$17.24_{\pm.72}$	$\mathbf{4.42}_{\pm.31}$	$\mathbf{6.67}_{\pm4.58}$	$19.08_{\pm1.15}$	$4.91_{\pm.37}$	$7.58_{\pm5.08}$	$20.54_{\pm1.81}$
	+ TAIF-N	$3.36_{\pm.49}$	$3.26_{\pm2.02}$	$15.41_{\pm.53}$	$4.14_{\pm.49}$	$5.56_{\pm3.65}$	$\mathbf{16.82}_{\pm.73}$	$5.12_{\pm.48}$	$6.99_{\pm4.74}$	$\mathbf{18.06}_{\pm.89}$	$5.63_{\pm.58}$	$7.73_{\pm5.06}$	$\mathbf{19.34}_{\pm1.22}$
DiffPuter	Baseline	$3.86_{\pm.75}$	$5.59_{\pm2.99}$	$16.25_{\pm.99}$	$4.46_{\pm.68}$	$\mathbf{6.32}_{\pm2.86}$	$\mathbf{17.15}_{\pm.79}$	$5.05_{\pm.63}$	$7.53_{\pm3.34}$	$18.08_{\pm.94}$	$5.45_{\pm.64}$	$\mathbf{8.46}_{\pm3.84}$	$\mathbf{18.94}_{\pm1.19}$
	+ TAIF-C	$\mathbf{3.82}_{\pm.69}$	$\mathbf{5.52}_{\pm3.02}$	$16.28_{\pm.94}$	$4.46_{\pm.55}$	$6.60_{\pm2.96}$	$17.30_{\pm.81}$	$5.09_{\pm.50}$	$7.76_{\pm3.43}$	$18.36_{\pm1.16}$	$5.63_{\pm.57}$	$8.65_{\pm3.96}$	$19.24_{\pm1.41}$
	+ TAIF-N	$3.90_{\pm.61}$	$8.76_{\pm8.59}$	$\mathbf{16.23}_{\pm1.00}$	$\mathbf{4.15}_{\pm.38}$	$7.42_{\pm5.68}$	$17.17_{\pm.80}$	$\mathbf{4.78}_{\pm.52}$	$\mathbf{7.07}_{\pm3.93}$	$\mathbf{18.07}_{\pm.93}$	$\mathbf{5.07}_{\pm.37}$	$8.55_{\pm4.46}$	$18.98_{\pm1.21}$

Table S8: Imputation performance (MAE ↓) on test set across different missing rates under **MAR** setting. Kin8nm values scaled by ×100. Best results within each base method in **bold**.

		Missing Rate = 30%			Missing Rate = 50%			Missing Rate = 70%			Missing Rate = 90%		
Base	Method	Hou.	Yac.	Kin.$_{\times 100}$	Hou.	Yac.	Kin.$_{\times 100}$	Hou.	Yac.	Kin.$_{\times 100}$	Hou.	Yac.	Kin.$_{\times 100}$
Mean	Baseline	**0.19**$_{\pm.01}$	**0.21**$_{\pm.03}$	**25.13**$_{\pm.19}$	**0.20**$_{\pm.01}$	**0.22**$_{\pm.02}$	**25.07**$_{\pm.20}$	**0.21**$_{\pm.01}$	**0.23**$_{\pm.02}$	**25.02**$_{\pm.17}$	**0.26**$_{\pm.02}$	**0.25**$_{\pm.03}$	**25.11**$_{\pm.18}$
	+ TAIF-C	0.19$_{\pm.01}$	0.21$_{\pm.03}$	25.13$_{\pm.19}$	0.20$_{\pm.01}$	0.22$_{\pm.02}$	25.07$_{\pm.20}$	0.21$_{\pm.01}$	0.23$_{\pm.02}$	25.02$_{\pm.17}$	0.26$_{\pm.02}$	0.25$_{\pm.03}$	25.11$_{\pm.18}$
	+ TAIF-N	1.32$_{\pm.16}$	1.30$_{\pm.39}$	25.31$_{\pm.25}$	1.45$_{\pm.07}$	1.43$_{\pm.47}$	25.20$_{\pm.22}$	1.69$_{\pm.22}$	1.94$_{\pm.50}$	25.22$_{\pm.22}$	2.63$_{\pm.39}$	2.54$_{\pm.72}$	26.33$_{\pm.41}$
Median	Baseline	**0.17**$_{\pm.02}$	**0.20**$_{\pm.02}$	**25.13**$_{\pm.19}$	**0.18**$_{\pm.01}$	**0.21**$_{\pm.01}$	**25.07**$_{\pm.21}$	**0.19**$_{\pm.01}$	**0.21**$_{\pm.02}$	**25.03**$_{\pm.18}$	**0.26**$_{\pm.03}$	**0.22**$_{\pm.03}$	**25.14**$_{\pm.18}$
	+ TAIF-C	0.17$_{\pm.02}$	0.20$_{\pm.02}$	25.13$_{\pm.19}$	0.18$_{\pm.01}$	0.21$_{\pm.01}$	25.07$_{\pm.21}$	0.19$_{\pm.01}$	0.21$_{\pm.02}$	25.03$_{\pm.18}$	0.26$_{\pm.03}$	0.22$_{\pm.03}$	25.14$_{\pm.18}$
	+ TAIF-N	1.36$_{\pm.24}$	1.38$_{\pm.39}$	25.31$_{\pm.23}$	1.52$_{\pm.23}$	1.73$_{\pm1.03}$	25.23$_{\pm.20}$	1.72$_{\pm.14}$	1.63$_{\pm.38}$	25.23$_{\pm.19}$	2.55$_{\pm.49}$	2.34$_{\pm.70}$	26.30$_{\pm.28}$
KNN	Baseline	**0.08**$_{\pm.01}$	**0.20**$_{\pm.02}$	26.77$_{\pm.14}$	0.09$_{\pm.01}$	**0.22**$_{\pm.02}$	26.80$_{\pm.27}$	0.11$_{\pm.01}$	**0.23**$_{\pm.02}$	26.79$_{\pm.22}$	0.16$_{\pm.01}$	**0.27**$_{\pm.03}$	26.83$_{\pm.15}$
	+ TAIF-C	0.08$_{\pm.01}$	0.21$_{\pm.02}$	26.93$_{\pm.27}$	**0.09**$_{\pm.01}$	0.23$_{\pm.01}$	26.76$_{\pm.18}$	**0.11**$_{\pm.00}$	0.24$_{\pm.02}$	26.73$_{\pm.17}$	**0.16**$_{\pm.01}$	0.27$_{\pm.03}$	26.77$_{\pm.19}$
	+ TAIF-N	1.37$_{\pm.19}$	1.45$_{\pm.62}$	**25.96**$_{\pm.19}$	1.45$_{\pm.14}$	1.64$_{\pm.80}$	**25.79**$_{\pm.19}$	1.66$_{\pm.12}$	1.87$_{\pm.54}$	**25.74**$_{\pm.20}$	2.54$_{\pm.67}$	2.14$_{\pm.65}$	**26.30**$_{\pm.24}$
MICE	Baseline	0.13$_{\pm.02}$	0.18$_{\pm.02}$	**25.13**$_{\pm.18}$	0.13$_{\pm.01}$	**0.21**$_{\pm.02}$	**26.21**$_{\pm1.70}$	0.15$_{\pm.01}$	**0.23**$_{\pm.02}$	27.07$_{\pm.40}$	0.24$_{\pm.02}$	**0.26**$_{\pm.02}$	**26.10**$_{\pm.32}$
	+ TAIF-C	**0.13**$_{\pm.01}$	**0.18**$_{\pm.02}$	25.50$_{\pm.20}$	**0.13**$_{\pm.01}$	0.22$_{\pm.02}$	27.37$_{\pm.39}$	**0.15**$_{\pm.01}$	0.23$_{\pm.03}$	**26.30**$_{\pm.34}$	**0.24**$_{\pm.02}$	0.28$_{\pm.03}$	26.17$_{\pm.25}$
	+ TAIF-N	1.33$_{\pm.21}$	1.49$_{\pm.71}$	25.32$_{\pm.23}$	1.47$_{\pm.28}$	1.74$_{\pm.63}$	26.26$_{\pm1.52}$	1.61$_{\pm.19}$	1.97$_{\pm.91}$	26.72$_{\pm.33}$	2.78$_{\pm.36}$	2.34$_{\pm.50}$	26.39$_{\pm.25}$
MissForest	Baseline	0.07$_{\pm.01}$	**0.14**$_{\pm.02}$	26.25$_{\pm.41}$	0.09$_{\pm.01}$	**0.18**$_{\pm.02}$	28.51$_{\pm.36}$	0.11$_{\pm.01}$	0.26$_{\pm.03}$	30.47$_{\pm.28}$	0.17$_{\pm.02}$	**0.32**$_{\pm.03}$	31.29$_{\pm.59}$
	+ TAIF-C	**0.07**$_{\pm.01}$	0.15$_{\pm.03}$	26.55$_{\pm.30}$	**0.08**$_{\pm.01}$	0.19$_{\pm.02}$	27.52$_{\pm.36}$	**0.11**$_{\pm.01}$	**0.26**$_{\pm.02}$	29.81$_{\pm.22}$	**0.16**$_{\pm.02}$	0.33$_{\pm.04}$	31.16$_{\pm.65}$
	+ TAIF-N	1.33$_{\pm.09}$	1.53$_{\pm.70}$	**25.92**$_{\pm.41}$	1.39$_{\pm.14}$	1.58$_{\pm.72}$	**25.89**$_{\pm.23}$	1.70$_{\pm.19}$	1.88$_{\pm.53}$	**25.90**$_{\pm.29}$	2.36$_{\pm.33}$	2.51$_{\pm.43}$	**26.87**$_{\pm.26}$
DiffPuter	Baseline	0.15$_{\pm.01}$	**0.21**$_{\pm.02}$	**25.35**$_{\pm.17}$	0.17$_{\pm.01}$	**0.22**$_{\pm.01}$	25.25$_{\pm.20}$	0.19$_{\pm.01}$	0.23$_{\pm.02}$	25.13$_{\pm.17}$	0.25$_{\pm.02}$	**0.25**$_{\pm.03}$	25.15$_{\pm.17}$
	+ TAIF-C	**0.15**$_{\pm.01}$	0.21$_{\pm.02}$	25.36$_{\pm.20}$	**0.17**$_{\pm.01}$	0.22$_{\pm.01}$	**25.22**$_{\pm.21}$	**0.19**$_{\pm.01}$	**0.23**$_{\pm.02}$	25.12$_{\pm.14}$	**0.25**$_{\pm.02}$	0.25$_{\pm.03}$	**25.15**$_{\pm.18}$
	+ TAIF-N	0.51$_{\pm.09}$	1.65$_{\pm.27}$	25.40$_{\pm.21}$	0.40$_{\pm.05}$	1.26$_{\pm.22}$	25.28$_{\pm.17}$	0.36$_{\pm.03}$	0.74$_{\pm.23}$	**25.12**$_{\pm.16}$	0.36$_{\pm.04}$	0.53$_{\pm.19}$	25.15$_{\pm.16}$

Table S9: Downstream prediction performance (MAE ↓) on test set across different missing rates under **MNAR (self-masking)** setting. Kin8nm values scaled by ×100. Best results within each base method in **bold**.

		Missing Rate = 30%			Missing Rate = 50%			Missing Rate = 70%			Missing Rate = 90%		
Base	Method	Hou.	Yac.	Kin.$_{\times 100}$	Hou.	Yac.	Kin.$_{\times 100}$	Hou.	Yac.	Kin.$_{\times 100}$	Hou.	Yac.	Kin.$_{\times 100}$
Mean	Baseline	**3.11**$_{\pm.31}$	2.17$_{\pm.43}$	15.92$_{\pm.23}$	**3.43**$_{\pm.41}$	3.92$_{\pm.52}$	18.16$_{\pm.30}$	**4.17**$_{\pm.34}$	6.70$_{\pm.63}$	**20.04**$_{\pm.34}$	**5.62**$_{\pm.49}$	**8.93**$_{\pm1.07}$	**21.46**$_{\pm.29}$
	+ TAIF-C	3.11$_{\pm.31}$	2.17$_{\pm.43}$	15.92$_{\pm.23}$	3.43$_{\pm.41}$	**3.92**$_{\pm.52}$	18.16$_{\pm.30}$	4.17$_{\pm.34}$	6.70$_{\pm.63}$	20.04$_{\pm.34}$	5.62$_{\pm.49}$	8.93$_{\pm1.07}$	21.46$_{\pm.29}$
	+ TAIF-N	3.39$_{\pm.25}$	**2.05**$_{\pm.53}$	**15.65**$_{\pm.29}$	4.21$_{\pm.32}$	4.17$_{\pm.60}$	**18.07**$_{\pm.32}$	5.01$_{\pm.40}$	**6.50**$_{\pm.63}$	20.35$_{\pm.39}$	6.09$_{\pm.49}$	9.16$_{\pm1.14}$	21.98$_{\pm.28}$
Median	Baseline	3.12$_{\pm.31}$	2.33$_{\pm.47}$	16.10$_{\pm.27}$	**3.47**$_{\pm.39}$	4.15$_{\pm.57}$	18.31$_{\pm.26}$	**4.22**$_{\pm.38}$	6.76$_{\pm.67}$	**20.09**$_{\pm.31}$	**5.63**$_{\pm.43}$	9.24$_{\pm1.22}$	**21.44**$_{\pm.28}$
	+ TAIF-C	**3.12**$_{\pm.31}$	2.33$_{\pm.47}$	16.10$_{\pm.27}$	3.47$_{\pm.39}$	4.15$_{\pm.57}$	18.31$_{\pm.26}$	4.22$_{\pm.38}$	6.76$_{\pm.67}$	20.09$_{\pm.31}$	5.63$_{\pm.43}$	9.24$_{\pm1.22}$	21.44$_{\pm.28}$
	+ TAIF-N	3.51$_{\pm.36}$	**2.19**$_{\pm.71}$	**15.62**$_{\pm.29}$	4.18$_{\pm.27}$	**4.06**$_{\pm.75}$	**18.12**$_{\pm.25}$	4.90$_{\pm.36}$	**6.58**$_{\pm.63}$	20.34$_{\pm.37}$	6.07$_{\pm.39}$	**9.09**$_{\pm.93}$	21.98$_{\pm.30}$
KNN	Baseline	3.38$_{\pm.42}$	2.48$_{\pm.38}$	17.70$_{\pm.24}$	4.16$_{\pm.33}$	4.82$_{\pm.71}$	19.64$_{\pm.35}$	4.87$_{\pm.51}$	6.94$_{\pm.85}$	20.86$_{\pm.31}$	5.98$_{\pm.56}$	9.57$_{\pm1.11}$	22.01$_{\pm.21}$
	+ TAIF-C	**3.00**$_{\pm.37}$	2.23$_{\pm.50}$	17.15$_{\pm.19}$	**3.47**$_{\pm.28}$	**3.96**$_{\pm.57}$	19.40$_{\pm.28}$	**4.36**$_{\pm.34}$	**6.52**$_{\pm.63}$	21.31$_{\pm.31}$	**5.84**$_{\pm.54}$	**9.05**$_{\pm1.21}$	22.75$_{\pm.38}$
	+ TAIF-N	3.35$_{\pm.23}$	**2.13**$_{\pm.43}$	**15.87**$_{\pm.21}$	4.35$_{\pm.16}$	4.18$_{\pm.68}$	**18.29**$_{\pm.32}$	4.94$_{\pm.37}$	7.08$_{\pm1.05}$	**20.10**$_{\pm.34}$	6.09$_{\pm.53}$	9.30$_{\pm1.05}$	**21.99**$_{\pm.27}$
MICE	Baseline	3.29$_{\pm.34}$	2.10$_{\pm.54}$	15.91$_{\pm.21}$	4.08$_{\pm.28}$	4.40$_{\pm.96}$	19.04$_{\pm.61}$	5.01$_{\pm.27}$	7.11$_{\pm.93}$	20.80$_{\pm.34}$	6.12$_{\pm.41}$	9.36$_{\pm1.29}$	22.11$_{\pm.31}$
	+ TAIF-C	**3.10**$_{\pm.32}$	1.98$_{\pm.44}$	16.77$_{\pm.22}$	**3.48**$_{\pm.24}$	**4.14**$_{\pm.70}$	19.40$_{\pm.32}$	**4.53**$_{\pm.32}$	**6.50**$_{\pm.79}$	21.73$_{\pm.25}$	**6.08**$_{\pm.49}$	**9.14**$_{\pm1.31}$	23.34$_{\pm.40}$
	+ TAIF-N	3.24$_{\pm.28}$	**1.97**$_{\pm.53}$	**15.53**$_{\pm.27}$	4.10$_{\pm.22}$	4.47$_{\pm.51}$	**18.40**$_{\pm.28}$	5.03$_{\pm.31}$	6.72$_{\pm.81}$	**20.57**$_{\pm.25}$	6.15$_{\pm.64}$	9.36$_{\pm1.15}$	**22.02**$_{\pm.30}$
MissForest	Baseline	3.06$_{\pm.39}$	2.27$_{\pm.43}$	17.38$_{\pm.26}$	3.89$_{\pm.24}$	4.18$_{\pm.92}$	19.82$_{\pm.34}$	4.88$_{\pm.29}$	6.83$_{\pm.86}$	20.99$_{\pm.25}$	6.33$_{\pm.21}$	9.70$_{\pm1.60}$	21.93$_{\pm.30}$
	+ TAIF-C	**2.84**$_{\pm.35}$	2.15$_{\pm.56}$	16.67$_{\pm.21}$	**3.45**$_{\pm.28}$	**4.05**$_{\pm.64}$	19.15$_{\pm.31}$	**4.19**$_{\pm.48}$	**6.55**$_{\pm.65}$	21.58$_{\pm.30}$	**5.87**$_{\pm.53}$	**9.10**$_{\pm1.30}$	22.90$_{\pm.32}$
	+ TAIF-N	3.22$_{\pm.32}$	**2.13**$_{\pm.49}$	**15.78**$_{\pm.25}$	4.24$_{\pm.47}$	4.59$_{\pm.90}$	**18.36**$_{\pm.21}$	5.06$_{\pm.37}$	6.85$_{\pm1.02}$	**20.23**$_{\pm.28}$	6.10$_{\pm.54}$	9.24$_{\pm.88}$	**21.82**$_{\pm.28}$
DiffPuter	Baseline	3.58$_{\pm.55}$	5.14$_{\pm3.05}$	17.65$_{\pm.83}$	4.13$_{\pm.53}$	6.27$_{\pm2.63}$	19.17$_{\pm.48}$	4.80$_{\pm.48}$	**7.97**$_{\pm1.74}$	**20.36**$_{\pm.37}$	**6.01**$_{\pm.51}$	10.06$_{\pm1.33}$	21.32$_{\pm.23}$
	+ TAIF-C	**3.47**$_{\pm.52}$	**4.82**$_{\pm2.89}$	17.64$_{\pm.68}$	4.02$_{\pm.45}$	**6.03**$_{\pm2.38}$	19.43$_{\pm.34}$	4.70$_{\pm.39}$	8.20$_{\pm1.57}$	20.94$_{\pm.67}$	6.03$_{\pm.50}$	10.42$_{\pm1.42}$	21.72$_{\pm.55}$
	+ TAIF-N	3.50$_{\pm.42}$	7.40$_{\pm6.63}$	**17.60**$_{\pm.87}$	**3.80**$_{\pm.32}$	9.82$_{\pm7.51}$	**19.15**$_{\pm.50}$	**4.56**$_{\pm.27}$	8.36$_{\pm2.93}$	20.39$_{\pm.31}$	6.03$_{\pm.65}$	**9.35**$_{\pm1.20}$	**21.30**$_{\pm.23}$

Table S10: Imputation performance (MAE ↓) on test set across different missing rates under **MNAR (self-masking)** setting. Kin8nm values scaled by ×100. Best results within each base method in **bold**.

		Missing Rate = 30%			Missing Rate = 50%			Missing Rate = 70%			Missing Rate = 90%		
Base	Method	Hou.	Yac.	Kin.$_{\times 100}$	Hou.	Yac.	Kin.$_{\times 100}$	Hou.	Yac.	Kin.$_{\times 100}$	Hou.	Yac.	Kin.$_{\times 100}$
Mean	Baseline	**0.24**$_{\pm.03}$	**0.28**$_{\pm.02}$	31.05$_{\pm.22}$	**0.25**$_{\pm.02}$	**0.29**$_{\pm.01}$	31.33$_{\pm.17}$	**0.29**$_{\pm.02}$	**0.31**$_{\pm.02}$	31.56$_{\pm.18}$	**0.39**$_{\pm.04}$	**0.44**$_{\pm.05}$	**31.51**$_{\pm.20}$
	+ TAIF-C	0.24$_{\pm.03}$	0.28$_{\pm.02}$	31.05$_{\pm.22}$	0.25$_{\pm.02}$	0.29$_{\pm.01}$	31.33$_{\pm.17}$	0.29$_{\pm.02}$	0.31$_{\pm.02}$	31.56$_{\pm.18}$	0.39$_{\pm.04}$	0.44$_{\pm.05}$	31.51$_{\pm.20}$
	+ TAIF-N	1.30$_{\pm.19}$	0.82$_{\pm.14}$	**24.73**$_{\pm.29}$	1.15$_{\pm.12}$	0.87$_{\pm.14}$	**27.14**$_{\pm.32}$	1.32$_{\pm.10}$	0.87$_{\pm.22}$	**29.24**$_{\pm.34}$	1.43$_{\pm.33}$	2.07$_{\pm.56}$	33.23$_{\pm1.23}$
Median	Baseline	**0.21**$_{\pm.04}$	**0.24**$_{\pm.03}$	32.86$_{\pm.20}$	**0.22**$_{\pm.02}$	**0.22**$_{\pm.01}$	34.01$_{\pm.19}$	**0.30**$_{\pm.02}$	**0.28**$_{\pm.03}$	34.73$_{\pm.20}$	**0.42**$_{\pm.06}$	**0.50**$_{\pm.06}$	34.98$_{\pm.22}$
	+ TAIF-C	0.21$_{\pm.04}$	0.24$_{\pm.03}$	32.86$_{\pm.20}$	0.22$_{\pm.02}$	0.22$_{\pm.01}$	34.01$_{\pm.19}$	0.30$_{\pm.02}$	0.28$_{\pm.03}$	34.73$_{\pm.20}$	0.42$_{\pm.06}$	0.50$_{\pm.06}$	34.98$_{\pm.22}$
	+ TAIF-N	1.29$_{\pm.11}$	0.91$_{\pm.12}$	**24.66**$_{\pm.29}$	1.20$_{\pm.17}$	0.85$_{\pm.11}$	**27.24**$_{\pm.41}$	1.24$_{\pm.15}$	0.97$_{\pm.18}$	**29.32**$_{\pm.46}$	1.27$_{\pm.13}$	1.84$_{\pm.49}$	**33.73**$_{\pm.72}$
KNN	Baseline	0.14$_{\pm.02}$	**0.26**$_{\pm.02}$	32.09$_{\pm.22}$	0.20$_{\pm.02}$	**0.27**$_{\pm.02}$	32.43$_{\pm.27}$	0.24$_{\pm.01}$	**0.31**$_{\pm.01}$	32.52$_{\pm.18}$	0.36$_{\pm.03}$	0.44$_{\pm.05}$	32.03$_{\pm.21}$
	+ TAIF-C	**0.13**$_{\pm.02}$	0.27$_{\pm.02}$	31.05$_{\pm.40}$	**0.15**$_{\pm.01}$	0.27$_{\pm.01}$	31.61$_{\pm.67}$	**0.19**$_{\pm.02}$	0.32$_{\pm.02}$	31.65$_{\pm.34}$	**0.29**$_{\pm.03}$	**0.42**$_{\pm.04}$	**31.78**$_{\pm.98}$
	+ TAIF-N	1.32$_{\pm.12}$	0.86$_{\pm.14}$	**25.40**$_{\pm.43}$	1.16$_{\pm.06}$	0.94$_{\pm.25}$	**28.03**$_{\pm.39}$	1.27$_{\pm.20}$	0.88$_{\pm.21}$	**29.72**$_{\pm.47}$	1.40$_{\pm.18}$	1.98$_{\pm.47}$	32.74$_{\pm.67}$
MICE	Baseline	0.17$_{\pm.02}$	0.21$_{\pm.03}$	31.05$_{\pm.22}$	0.18$_{\pm.01}$	0.26$_{\pm.02}$	32.69$_{\pm1.01}$	0.24$_{\pm.03}$	0.31$_{\pm.01}$	32.76$_{\pm.27}$	0.39$_{\pm.05}$	0.47$_{\pm.05}$	32.20$_{\pm.24}$
	+ TAIF-C	**0.17**$_{\pm.02}$	**0.20**$_{\pm.02}$	30.93$_{\pm.43}$	**0.17**$_{\pm.01}$	**0.24**$_{\pm.01}$	31.04$_{\pm.22}$	**0.23**$_{\pm.02}$	**0.30**$_{\pm.01}$	31.24$_{\pm.12}$	**0.38**$_{\pm.05}$	**0.44**$_{\pm.05}$	**31.51**$_{\pm.28}$
	+ TAIF-N	1.17$_{\pm.12}$	0.83$_{\pm.16}$	**24.69**$_{\pm.47}$	1.15$_{\pm.12}$	0.98$_{\pm.20}$	**28.90**$_{\pm1.19}$	1.22$_{\pm.15}$	0.81$_{\pm.19}$	**30.47**$_{\pm.52}$	1.35$_{\pm.16}$	1.95$_{\pm.38}$	31.68$_{\pm.86}$
MissForest	Baseline	**0.10**$_{\pm.02}$	0.18$_{\pm.03}$	31.69$_{\pm.18}$	0.12$_{\pm.01}$	0.22$_{\pm.02}$	33.45$_{\pm.40}$	0.17$_{\pm.02}$	**0.29**$_{\pm.03}$	34.81$_{\pm.37}$	0.32$_{\pm.04}$	0.47$_{\pm.05}$	34.12$_{\pm.78}$
	+ TAIF-C	0.10$_{\pm.02}$	**0.17**$_{\pm.02}$	30.13$_{\pm.40}$	**0.12**$_{\pm.01}$	**0.21**$_{\pm.02}$	31.30$_{\pm.42}$	**0.16**$_{\pm.02}$	0.29$_{\pm.02}$	32.85$_{\pm.30}$	**0.27**$_{\pm.04}$	**0.45**$_{\pm.04}$	34.30$_{\pm.95}$
	+ TAIF-N	1.26$_{\pm.10}$	0.93$_{\pm.15}$	**25.08**$_{\pm.41}$	1.10$_{\pm.09}$	0.93$_{\pm.13}$	**28.08**$_{\pm.45}$	1.21$_{\pm.13}$	0.97$_{\pm.17}$	**30.63**$_{\pm.47}$	1.27$_{\pm.20}$	2.10$_{\pm.37}$	**33.11**$_{\pm.56}$
DiffPuter	Baseline	0.20$_{\pm.03}$	0.28$_{\pm.02}$	31.29$_{\pm.36}$	0.23$_{\pm.01}$	0.28$_{\pm.01}$	31.47$_{\pm.20}$	0.28$_{\pm.02}$	0.31$_{\pm.02}$	31.63$_{\pm.17}$	0.38$_{\pm.04}$	0.44$_{\pm.05}$	31.50$_{\pm.19}$
	+ TAIF-C	**0.20**$_{\pm.03}$	**0.27**$_{\pm.02}$	**31.05**$_{\pm.23}$	**0.23**$_{\pm.01}$	**0.28**$_{\pm.01}$	**31.32**$_{\pm.17}$	**0.27**$_{\pm.02}$	**0.31**$_{\pm.02}$	**31.56**$_{\pm.19}$	**0.38**$_{\pm.04}$	**0.44**$_{\pm.05}$	**31.46**$_{\pm.20}$
	+ TAIF-N	0.51$_{\pm.06}$	1.47$_{\pm.19}$	31.17$_{\pm.28}$	0.41$_{\pm.04}$	1.28$_{\pm.19}$	31.45$_{\pm.17}$	0.38$_{\pm.02}$	0.81$_{\pm.21}$	31.61$_{\pm.18}$	0.42$_{\pm.04}$	0.52$_{\pm.05}$	31.50$_{\pm.20}$

Table S11: Downstream prediction performance (MAE ↓) on test set across different missing rates under **MNAR** setting. Kin8nm values scaled by ×100. Best results within each base method in **bold**.

		Missing Rate = 30%			Missing Rate = 50%			Missing Rate = 70%			Missing Rate = 90%		
Base	Method	Hou.	Yac.	Kin.$_{\times 100}$	Hou.	Yac.	Kin.$_{\times 100}$	Hou.	Yac.	Kin.$_{\times 100}$	Hou.	Yac.	Kin.$_{\times 100}$
Mean	Baseline	**3.43**$_{\pm.42}$	**4.14**$_{\pm.76}$	16.56$_{\pm.21}$	**4.10**$_{\pm.34}$	**6.47**$_{\pm.91}$	18.17$_{\pm.18}$	**5.20**$_{\pm.33}$	8.77$_{\pm1.51}$	**19.60**$_{\pm.27}$	**6.59**$_{\pm.40}$	9.99$_{\pm1.55}$	**21.16**$_{\pm.20}$
	+ TAIF-C	3.43$_{\pm.42}$	4.14$_{\pm.76}$	16.56$_{\pm.21}$	4.10$_{\pm.34}$	6.47$_{\pm.91}$	18.17$_{\pm.18}$	5.20$_{\pm.33}$	8.77$_{\pm1.51}$	19.60$_{\pm.27}$	6.59$_{\pm.40}$	9.99$_{\pm1.55}$	21.16$_{\pm.20}$
	+ TAIF-N	4.00$_{\pm.58}$	4.58$_{\pm.99}$	**16.38**$_{\pm.26}$	5.02$_{\pm.51}$	6.52$_{\pm1.14}$	**18.05**$_{\pm.18}$	5.72$_{\pm.26}$	**8.49**$_{\pm1.69}$	19.93$_{\pm.34}$	6.71$_{\pm.60}$	**9.98**$_{\pm1.36}$	21.57$_{\pm.17}$
Median	Baseline	**3.45**$_{\pm.43}$	**4.33**$_{\pm.84}$	16.57$_{\pm.22}$	**4.00**$_{\pm.22}$	**6.62**$_{\pm.83}$	18.16$_{\pm.19}$	**5.10**$_{\pm.24}$	8.81$_{\pm1.46}$	**19.60**$_{\pm.28}$	**6.59**$_{\pm.39}$	**9.81**$_{\pm1.66}$	**21.14**$_{\pm.19}$
	+ TAIF-C	3.45$_{\pm.43}$	4.33$_{\pm.84}$	16.57$_{\pm.22}$	4.00$_{\pm.22}$	6.62$_{\pm.83}$	18.16$_{\pm.19}$	5.10$_{\pm.24}$	8.81$_{\pm1.46}$	19.60$_{\pm.28}$	6.59$_{\pm.39}$	9.81$_{\pm1.66}$	21.14$_{\pm.19}$
	+ TAIF-N	4.05$_{\pm.54}$	4.71$_{\pm.89}$	**16.31**$_{\pm.24}$	4.92$_{\pm.45}$	6.65$_{\pm1.16}$	**18.05**$_{\pm.21}$	5.79$_{\pm.37}$	**8.40**$_{\pm1.74}$	19.90$_{\pm.33}$	6.80$_{\pm.58}$	9.91$_{\pm1.35}$	21.56$_{\pm.17}$
KNN	Baseline	3.35$_{\pm.46}$	4.71$_{\pm.85}$	17.46$_{\pm.25}$	4.33$_{\pm.40}$	7.00$_{\pm.79}$	19.08$_{\pm.16}$	5.64$_{\pm.35}$	8.89$_{\pm1.23}$	20.16$_{\pm.24}$	**6.57**$_{\pm.37}$	**9.84**$_{\pm1.45}$	**21.61**$_{\pm.20}$
	+ TAIF-C	**3.06**$_{\pm.39}$	**4.26**$_{\pm.85}$	17.31$_{\pm.24}$	**3.77**$_{\pm.40}$	**6.38**$_{\pm1.01}$	19.36$_{\pm.22}$	**5.09**$_{\pm.29}$	**8.39**$_{\pm1.41}$	20.75$_{\pm.29}$	6.63$_{\pm.52}$	10.33$_{\pm1.25}$	22.43$_{\pm.37}$
	+ TAIF-N	3.81$_{\pm.66}$	4.94$_{\pm1.04}$	**16.55**$_{\pm.30}$	4.75$_{\pm.52}$	6.67$_{\pm1.42}$	**18.18**$_{\pm.22}$	5.87$_{\pm.52}$	8.92$_{\pm1.73}$	**19.57**$_{\pm.30}$	6.88$_{\pm.50}$	9.99$_{\pm1.41}$	21.64$_{\pm.20}$
MICE	Baseline	3.52$_{\pm.45}$	4.78$_{\pm1.23}$	16.52$_{\pm.21}$	4.26$_{\pm.41}$	6.87$_{\pm.69}$	18.46$_{\pm.35}$	5.55$_{\pm.39}$	8.95$_{\pm1.63}$	20.12$_{\pm.25}$	6.82$_{\pm.50}$	10.09$_{\pm1.40}$	21.68$_{\pm.24}$
	+ TAIF-C	**3.40**$_{\pm.40}$	**4.09**$_{\pm.65}$	17.06$_{\pm.21}$	**4.00**$_{\pm.38}$	**6.38**$_{\pm.66}$	19.19$_{\pm.19}$	**5.39**$_{\pm.32}$	8.66$_{\pm1.61}$	21.23$_{\pm.32}$	**6.65**$_{\pm.47}$	10.22$_{\pm1.39}$	22.97$_{\pm.38}$
	+ TAIF-N	3.88$_{\pm.47}$	4.54$_{\pm1.03}$	**16.33**$_{\pm.20}$	4.80$_{\pm.30}$	6.78$_{\pm1.44}$	**18.20**$_{\pm.22}$	5.77$_{\pm.36}$	**8.58**$_{\pm1.59}$	**20.04**$_{\pm.23}$	6.96$_{\pm.61}$	**9.91**$_{\pm1.46}$	**21.67**$_{\pm.25}$
MissForest	Baseline	3.27$_{\pm.34}$	4.36$_{\pm.85}$	16.80$_{\pm.24}$	4.10$_{\pm.46}$	7.06$_{\pm1.15}$	18.99$_{\pm.24}$	5.54$_{\pm.36}$	8.70$_{\pm1.56}$	20.57$_{\pm.20}$	**6.68**$_{\pm.39}$	10.14$_{\pm.99}$	21.78$_{\pm.15}$
	+ TAIF-C	**3.09**$_{\pm.34}$	**4.21**$_{\pm.74}$	16.75$_{\pm.21}$	**3.82**$_{\pm.46}$	6.55$_{\pm1.12}$	18.82$_{\pm.26}$	**5.30**$_{\pm.38}$	**8.58**$_{\pm1.60}$	20.77$_{\pm.37}$	6.80$_{\pm.42}$	10.09$_{\pm1.07}$	22.52$_{\pm.33}$
	+ TAIF-N	3.74$_{\pm.43}$	4.90$_{\pm.84}$	**16.47**$_{\pm.26}$	4.68$_{\pm.47}$	**6.52**$_{\pm1.30}$	**18.27**$_{\pm.22}$	5.70$_{\pm.27}$	8.59$_{\pm1.87}$	**19.71**$_{\pm.34}$	6.84$_{\pm.53}$	**9.84**$_{\pm1.40}$	**21.65**$_{\pm.25}$
DiffPuter	Baseline	3.98$_{\pm.69}$	**6.40**$_{\pm2.27}$	**17.32**$_{\pm.75}$	4.65$_{\pm.58}$	**7.61**$_{\pm1.23}$	18.64$_{\pm.41}$	**5.53**$_{\pm.36}$	9.20$_{\pm1.06}$	**19.76**$_{\pm.30}$	**6.67**$_{\pm.46}$	**10.30**$_{\pm.99}$	21.07$_{\pm.18}$
	+ TAIF-C	3.86$_{\pm.65}$	6.43$_{\pm2.24}$	17.44$_{\pm.57}$	4.68$_{\pm.47}$	7.87$_{\pm1.32}$	18.99$_{\pm.30}$	5.64$_{\pm.30}$	9.39$_{\pm1.27}$	20.32$_{\pm.55}$	6.73$_{\pm.49}$	10.49$_{\pm1.02}$	21.36$_{\pm.39}$
	+ TAIF-N	**3.86**$_{\pm.51}$	13.92$_{\pm11.78}$	17.34$_{\pm.70}$	**4.28**$_{\pm.50}$	11.58$_{\pm6.14}$	**18.62**$_{\pm.42}$	5.63$_{\pm.42}$	**8.70**$_{\pm1.40}$	19.78$_{\pm.28}$	7.12$_{\pm.62}$	10.35$_{\pm1.28}$	**21.07**$_{\pm.21}$

Table S12: Imputation performance (MAE ↓) on test set across different missing rates under **MNAR** setting. Kin8nm values scaled by ×100. Best results within each base method in **bold**.

		Missing Rate = 30%			Missing Rate = 50%			Missing Rate = 70%			Missing Rate = 90%		
Base	Method	Hou.	Yac.	Kin.$_{\times 100}$	Hou.	Yac.	Kin.$_{\times 100}$	Hou.	Yac.	Kin.$_{\times 100}$	Hou.	Yac.	Kin.$_{\times 100}$
Mean	Baseline	**0.19**$_{\pm.01}$	**0.21**$_{\pm.01}$	25.09$_{\pm.22}$	**0.20**$_{\pm.01}$	**0.22**$_{\pm.01}$	25.01$_{\pm.19}$	**0.21**$_{\pm.01}$	**0.22**$_{\pm.01}$	24.99$_{\pm.17}$	**0.25**$_{\pm.02}$	**0.24**$_{\pm.02}$	**25.08**$_{\pm.15}$
	+ TAIF-C	0.19$_{\pm.01}$	0.21$_{\pm.01}$	25.09$_{\pm.22}$	0.20$_{\pm.01}$	0.22$_{\pm.01}$	25.01$_{\pm.19}$	0.21$_{\pm.01}$	0.22$_{\pm.01}$	24.99$_{\pm.17}$	0.25$_{\pm.02}$	0.24$_{\pm.02}$	25.08$_{\pm.15}$
	+ TAIF-N	1.22$_{\pm.17}$	1.18$_{\pm.42}$	**24.77**$_{\pm.25}$	1.19$_{\pm.10}$	1.60$_{\pm.75}$	**24.55**$_{\pm.21}$	1.23$_{\pm.12}$	1.67$_{\pm.57}$	**24.72**$_{\pm.20}$	1.30$_{\pm.11}$	1.86$_{\pm.40}$	26.69$_{\pm.81}$
Median	Baseline	**0.17**$_{\pm.02}$	**0.19**$_{\pm.01}$	25.10$_{\pm.22}$	**0.17**$_{\pm.01}$	**0.21**$_{\pm.01}$	25.02$_{\pm.19}$	**0.19**$_{\pm.01}$	**0.21**$_{\pm.01}$	24.99$_{\pm.17}$	**0.25**$_{\pm.03}$	**0.22**$_{\pm.02}$	**25.10**$_{\pm.15}$
	+ TAIF-C	0.17$_{\pm.02}$	0.19$_{\pm.01}$	25.10$_{\pm.22}$	0.17$_{\pm.01}$	0.21$_{\pm.01}$	25.02$_{\pm.19}$	0.19$_{\pm.01}$	0.21$_{\pm.01}$	24.99$_{\pm.17}$	0.25$_{\pm.03}$	0.22$_{\pm.02}$	25.10$_{\pm.15}$
	+ TAIF-N	1.25$_{\pm.22}$	1.23$_{\pm.43}$	**24.72**$_{\pm.23}$	1.29$_{\pm.15}$	1.27$_{\pm.58}$	**24.57**$_{\pm.17}$	1.23$_{\pm.16}$	1.58$_{\pm.46}$	**24.66**$_{\pm.15}$	1.32$_{\pm.23}$	2.16$_{\pm.20}$	26.65$_{\pm.56}$
KNN	Baseline	0.10$_{\pm.01}$	**0.21**$_{\pm.03}$	28.23$_{\pm.33}$	0.15$_{\pm.01}$	**0.22**$_{\pm.02}$	27.39$_{\pm.15}$	0.17$_{\pm.01}$	**0.24**$_{\pm.02}$	26.30$_{\pm.15}$	0.24$_{\pm.02}$	**0.26**$_{\pm.02}$	25.30$_{\pm.14}$
	+ TAIF-C	**0.09**$_{\pm.01}$	0.22$_{\pm.02}$	28.15$_{\pm.25}$	**0.12**$_{\pm.01}$	0.25$_{\pm.02}$	28.20$_{\pm.26}$	**0.14**$_{\pm.01}$	0.26$_{\pm.02}$	27.37$_{\pm.28}$	**0.21**$_{\pm.02}$	0.27$_{\pm.03}$	26.91$_{\pm.39}$
	+ TAIF-N	1.19$_{\pm.14}$	1.15$_{\pm.36}$	**25.60**$_{\pm.28}$	1.16$_{\pm.14}$	1.48$_{\pm.61}$	**25.09**$_{\pm.26}$	1.16$_{\pm.10}$	1.47$_{\pm.54}$	**24.93**$_{\pm.17}$	1.34$_{\pm.08}$	1.97$_{\pm.58}$	**25.19**$_{\pm.21}$
MICE	Baseline	0.14$_{\pm.01}$	0.19$_{\pm.02}$	25.09$_{\pm.21}$	0.14$_{\pm.01}$	**0.22**$_{\pm.01}$	26.96$_{\pm1.61}$	0.18$_{\pm.01}$	**0.23**$_{\pm.02}$	26.78$_{\pm.31}$	0.27$_{\pm.03}$	0.29$_{\pm.04}$	26.19$_{\pm.31}$
	+ TAIF-C	**0.14**$_{\pm.01}$	**0.18**$_{\pm.01}$	26.30$_{\pm.98}$	**0.14**$_{\pm.01}$	0.22$_{\pm.01}$	26.11$_{\pm.23}$	**0.18**$_{\pm.01}$	0.24$_{\pm.03}$	**25.58**$_{\pm.20}$	**0.27**$_{\pm.04}$	**0.28**$_{\pm.03}$	25.89$_{\pm.25}$
	+ TAIF-N	1.15$_{\pm.19}$	1.13$_{\pm.41}$	**24.72**$_{\pm.24}$	1.27$_{\pm.21}$	1.45$_{\pm.72}$	**25.67**$_{\pm.95}$	1.25$_{\pm.16}$	1.76$_{\pm.88}$	25.99$_{\pm.31}$	1.27$_{\pm.12}$	2.23$_{\pm.44}$	**25.86**$_{\pm.28}$
MissForest	Baseline	0.08$_{\pm.01}$	0.15$_{\pm.01}$	25.57$_{\pm.34}$	0.10$_{\pm.01}$	0.20$_{\pm.02}$	27.29$_{\pm.32}$	0.13$_{\pm.01}$	**0.26**$_{\pm.03}$	29.26$_{\pm.32}$	0.25$_{\pm.02}$	0.32$_{\pm.04}$	30.01$_{\pm1.06}$
	+ TAIF-C	**0.08**$_{\pm.01}$	**0.15**$_{\pm.02}$	26.12$_{\pm.32}$	**0.10**$_{\pm.01}$	**0.19**$_{\pm.01}$	26.97$_{\pm.30}$	**0.12**$_{\pm.01}$	0.26$_{\pm.03}$	28.09$_{\pm.34}$	**0.22**$_{\pm.01}$	**0.31**$_{\pm.05}$	29.74$_{\pm1.01}$
	+ TAIF-N	1.23$_{\pm.21}$	1.08$_{\pm.29}$	**25.05**$_{\pm.32}$	1.27$_{\pm.14}$	1.20$_{\pm.42}$	**25.07**$_{\pm.15}$	1.23$_{\pm.16}$	1.54$_{\pm.46}$	**25.29**$_{\pm.12}$	1.17$_{\pm.21}$	1.80$_{\pm.43}$	**26.20**$_{\pm.49}$
DiffPuter	Baseline	0.15$_{\pm.01}$	0.21$_{\pm.01}$	25.35$_{\pm.21}$	0.17$_{\pm.01}$	0.22$_{\pm.01}$	25.12$_{\pm.17}$	0.20$_{\pm.01}$	0.23$_{\pm.01}$	25.02$_{\pm.15}$	0.25$_{\pm.02}$	0.24$_{\pm.02}$	25.10$_{\pm.14}$
	+ TAIF-C	**0.15**$_{\pm.01}$	**0.21**$_{\pm.01}$	**25.30**$_{\pm.20}$	**0.17**$_{\pm.01}$	**0.22**$_{\pm.01}$	**25.08**$_{\pm.19}$	**0.20**$_{\pm.01}$	**0.22**$_{\pm.01}$	**25.00**$_{\pm.15}$	**0.25**$_{\pm.02}$	**0.24**$_{\pm.02}$	**25.09**$_{\pm.15}$
	+ TAIF-N	0.45$_{\pm.05}$	1.74$_{\pm.20}$	25.36$_{\pm.22}$	0.35$_{\pm.04}$	1.20$_{\pm.23}$	25.13$_{\pm.21}$	0.31$_{\pm.03}$	0.57$_{\pm.08}$	25.02$_{\pm.16}$	0.30$_{\pm.02}$	0.37$_{\pm.05}$	25.09$_{\pm.15}$

4 Performance Across Missing Rates

Figures S1 and S2 visualize performance across missing rates. For each dataset, we select the best-performing preliminary imputer and compare with its TAIF-enhanced variant. The shaded area indicates improvement from TAIF. TAIF consistently improves performance across both classification and regression tasks.

Figure S1: Classification accuracy (%) across missing rates. Higher is better.

Figure S2: Regression MAE across missing rates. Lower is better.

5 Pseudo-Label Method Ablation

Table S13 presents an ablation study on the choice of pseudo-label generation method in Stage 1 of TAIF-C. We compare four strategies: using a single imputation method (Mean, MICE, or MissForest) versus an ensemble approach that averages predictions from all three methods.

Across all three datasets and five base imputers, the ensemble approach consistently achieves the lowest average rank. This suggests that combining multiple imputation methods for pseudo-label generation reduces the bias introduced by any single method's inductive assumptions. Notably, the benefit of ensemble pseudo-labels is most pronounced for weaker base imputers such as KNN, where the average rank improves substantially compared to single-method alternatives.

Among single-method approaches, no method dominates across all settings. Mean imputation performs competitively on Energy and Yacht despite its simplicity, while MICE shows strength on Housing. This variability reinforces the robustness advantage of the ensemble strategy, which consistently avoids worst-case performance while achieving the best average rank across all datasets.

Table S13: Ablation on pseudo-label methods for TAIF-C. Average rank (↓) across datasets and base imputers. Best per column in **bold**.

S1 Method	KNN	MICE	MissForest	GAIN	Diffputer	Avg
			Energy			
Mean	2.68	2.32	**2.36**	2.67	2.50	2.51
MICE	2.70	2.60	2.74	2.92	2.52	2.70
MissForest	2.63	2.88	2.38	**2.21**	**2.46**	2.51
Ensemble	**1.98**	**2.21**	2.51	**2.21**	2.52	**2.29**
			Housing			
Mean	2.74	2.61	2.92	3.37	**2.30**	2.79
MICE	2.30	2.51	2.44	2.16	2.61	2.40
MissForest	2.91	2.72	2.51	2.19	2.54	2.58
Ensemble	**2.05**	**2.16**	**2.13**	**2.14**	2.56	**2.21**
			Yacht			
Mean	2.70	**2.32**	2.44	2.38	2.74	2.52
MICE	2.60	2.78	2.69	3.00	2.52	2.72
MissForest	2.37	2.45	2.46	**2.31**	2.53	2.42
Ensemble	**2.33**	2.45	**2.41**	**2.31**	**2.21**	**2.34**

6 Decoupling Phenomenon

Table S14 reports the Spearman correlation between imputation error and downstream prediction error for each dataset and missing rate combination. This table provides the numerical values corresponding to Figure 6 in the main paper in the main text.

Fewer than half of the configurations exhibit statistically significant correlation ($p < 0.05$), and the overall average correlation of 0.46 indicates only a moderate relationship. Notably, Protein shows consistently weak correlations across all missing rates, with an average of only 0.07. Yacht at 50% missing rate exhibits a negative correlation of -0.57, indicating that better reconstruction actually harms prediction in this setting.

Table S14: Spearman correlation (ρ) between imputation error and downstream prediction error across methods. Values with $p < 0.05$ are marked with *. Weak or insignificant correlations indicate that reconstruction quality fails to predict downstream performance.

Dataset	30%	50%	70%	90%	Avg.
Housing	0.62^*	0.45	0.43	0.80^*	0.57
Energy	0.72^*	0.43	0.58^*	0.71^*	0.61
Yacht	0.32	-0.57	0.57	0.65^*	0.24
Kin8nm	0.80^*	0.76^*	0.79^*	0.48	0.71
Power	0.40	0.40	0.56	0.88^*	0.56
Protein	0.27	0.20	0.15	-0.32	0.07
Avg.	0.52	0.28	0.51	0.53	0.46

7 Computational Overhead

Table S15 reports detailed timing analysis for TAIF with deep learning baselines. We decompose total runtime into Stage 1 (task signal generation) and Stage 2 (imputation) for different pseudo-label methods.

For most configurations, Stage 1 accounts for less than 8% of total runtime. Even with ensemble-based pseudo-labels, which yield the robust downstream performance, Stage 1 overhead remains below 8% for DiffPuter and below 3% for GRAPE. This is because MissForest dominates the ensemble computation time. These results demonstrate that TAIF achieves task-aware imputation with negligible computational overhead relative to the base imputation methods.

Table S15: TAIF timing analysis at 30% missing rate. S1 and S2 denote Stage 1 and Stage 2 runtime in seconds.

Model	**Dataset (Pseudo)**	**S1 (s)**	**S2 (s)**	**S1 (%)**
DiffPuter	Energy (Mean)	0.3	43.4	0.8
	Energy (MICE)	0.6	42.7	1.3
	Energy (MissForest)	1.8	36.8	4.7
	Energy (Ensemble)	2.3	34.7	6.2
	Yacht (Mean)	0.2	18.8	1.1
	Yacht (MICE)	0.3	20.0	1.5
	Yacht (MissForest)	1.1	19.7	5.2
	Yacht (Ensemble)	1.6	19.7	7.5
GRAPE	Energy (Mean)	0.2	168.7	0.1
	Energy (MICE)	0.3	169.1	0.2
	Energy (MissForest)	4.3	167.4	2.5
	Energy (Ensemble)	4.8	168.4	2.8
	Yacht (Mean)	0.2	251.3	0.1
	Yacht (MICE)	0.3	240.8	0.1
	Yacht (MissForest)	3.8	238.6	1.6
	Yacht (Ensemble)	4.0	222.1	1.8

References